FUNDAMENTALS OF INVESTMENTS
Lecture notes

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Forewords

These lecture notes contain the condensed core readings for the **FIN-A0104 – Fundamentals of Investments** course at Aalto University School of Business.

A typical introductory investments textbook is hundreds, if not over a thousand, pages long. These lecture notes are an attempt to condense the central topics relevant for the course in a very small amount of pages. The natural consequence is that this document contains less explanations and examples than a traditional textbook. A reader looking for a more thorough treatment of any of the topics covered here will be well served by any edition of **Investments** by Bodie, Kane, and Marcus.

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Part I

Fundamentals

Chapter 1

Assets and markets

This chapter introduces the basic concepts and terminology of investing and financial assets and markets.

1.1 Investing, assets and securities

Investing is the process of committing resources today for the acquisition of **assets** that are expected to generate some benefits in the future. For example, this can be buying stocks today to enjoy dividends and price appreciation in the future, or sacrificing time and money today on education in order to enjoy higher lifetime earnings and greater job satisfaction in the future.

Assets can be divided into two broad categories. **Real assets** are things like land, buildings, machinery, and knowledge that we use to generate the net income of our economy. **Financial assets** are things like stocks and bonds. These are financial contracts about who gets what share of the income generated by the real assets and who controls the real assets. As such, financial assets do not generate income. However, they facilitate the accumulation of the income-generating real assets.

Securities are financial assets that are tradable and fungible. Securities can be

exchanged easily for cash or other financial securities. Historically, securities were in the form of a paper certificate, but nowadays they mainly take the form of a digital book entry. The two main categories of financial securities are debt and equity.

Debt securities or **fixed income securities** are a form of securitized borrowing. The borrower (for example a government or a firm) borrows money from a lender by issuing a debt security. The lender gives money to the borrower and receives the debt security in return. The borrower promises to make fixed payments to the holder of the security: periodic coupon payments and par value at maturity. The lender can either hold the security until maturity or sell it to another investor who then inherits the rights to receive the payments. Debt securities typically do not come with any control rights over the assets of the issuer.

The market for fixed income securities is divided in two segments. The market for debt securities with one year or less to maturity is known as **money market**. Money market securities issued by governments are called **government bills**, whereas those issued by banks are called **certificates of deposit** and those issued by firms are known as **commercial paper**. Fixed income securities with more than one year to maturity are known as **bonds**. Sometimes bonds with less than ten years to maturity are called **notes**.

There are two main sources of risk associated with investing in fixed income securities, interest rate risk and credit risk. **Interest rate risk** refers to the fact that when interest rates rise, the value of fixed income securities declines. Typically, the longer the maturity of the security, the greater the risk. **Credit risk** refers to the risk that the issuer will not have sufficient funds available to pay the coupons or the par value when due. Credit risk is typically relatively low in government bonds but varies widely across issuers in corporate bonds. Chapters 7 and 8 provide a more detailed analysis of fixed income investments.

Equity securities are basically common stocks of corporations. They provide their holders two kinds of rights. **Cash flow rights** refer to the equityholders' right to receive, as **dividends**, their share of the firm's profits. Note that the

amount of dividend is not fixed, can vary substantially over time, and can also be zero for long periods of time. Equityholders also control the firm; this is known as **control rights**. Equityholders exercise this control by appointing the board of directors and by voting on various issues brought to the general meeting of the shareholders.

Indexes make tracking the overall development of the market, or a segment of the market, easier. An index is basically a basket of securities chosen following a predetermined rule. There are bond indexes tracking the developments of bond markets and there are equity indexes tracking the development of equity markets. Under the umbrella of equity indexes, there are global indexes, regional indexes, sector indexes, style indexes, and so on.

1.2 Financial markets

Financial markets are markets for financial securities. Buyers and sellers of financial securities meet in financial markets to exchange securities and cash under predetermined rules.

Financial markets can be seen to serve a number of roles in a well-functioning economy:

- **Facilitation of trade**. Financial markets match the demand for securities by buyers with the supply of securities by sellers. This results in transactions (i.e. securities changing ownership) as well as prices for the securities.
- **Production of information**. Investors seeking to maximize the investment profits have strong incentives to buy securities when they are too cheap and sell when too expensive. This results in the prices reflecting the future prospects of the issuer. By facilitating efficient trade in financial securities, financial markets ensure that the price signals are as informative as possible.
- **Allocation of resources**. Financial markets, and market participants, use the information in prices to allocate resources to their highest valued uses.

- **Timing of consumption**. Investors can use the financial markets to move consumption in time. An individual with surplus resources now, can invest in financial securities and use the proceeds from these investments to finance consumption in the future when she has less resources otherwise. Saving for retirement is an example of this.

- **Sharing of risks**. Through well-functioning financial markets, investors and entrepreneurs can diversify their risks so that they hold multiple securities rather than just a few.

- **Separation of ownership and control**. Some individuals may be very good at managing firms but do not possess the financial resources to start a firm. Others may have the financial resources to finance a firm but not the skills to manage one. Financial markets enable the first group to seek financing for their firm from the second group ensuring that every individual in the economy does what they are best at.

Financial markets are typically thought to be very competitive. There are large amounts of investors seeking the best possible investments. Due to the potentially huge financial rewards, a lot of talent and resources are dedicated to identifying those best investments. This has the important implication that we are very unlikely to find free lunches on the financial markets. Due to the competitive market, prices of financial securities are likely to reflect all relevant information. Thus, finding too cheap or too expensive securities is impossible. As a result, expected profits only depend on the risks taken in investing, not on any informational advantages. We discuss this concept of efficient markets hypothesis in more detail in Section 2.4.

Financial markets have two sub-markets, primary and secondary. The term **primary market** is used to denote when securities (like stocks and bonds) are initially created and sold. In primary markets, issuer (like firms and governments) issue, or sell, the securities to investor. Investors receive the securities and the issuer receives the money. The primary market is where issuers raise money from the investors. The **secondary market** is where investors trade already issued secu-

rities among themselves. In the secondary market, securities and cash move from investors to other investors. The issuer is not a counterparty to transactions in the secondary market.[1] While the issuer does not directly benefit from the transactions in the secondary market, an active and liquid secondary market facilitates the functioning of the primary market. Thus, issuer benefit indirectly from the secondary market activity.

The main participants in the financial markets are financial intermediaries, households who are suppliers of capital (investors), and firms and governments who are net demanders of capital (issuers). Some firms and governments also invest through financial markets but on, aggregate, these sectors are net demanders. **Financial intermediaries** are institutions that help bring the suppliers and demanders of capital together in the market, such as commercial banks, investment banks, investment companies, investment funds, pension funds, brokers, financial advisers, and securities exchanges.

1.3 Trading financial securities

Broadly speaking, markets in general can be classified to be of one of four different types:

- In a **direct search market**, buyers and sellers find each other directly without a formalized market mechanism.
- In a **brokered market**, a broker is used to find and match buyers and sellers. Primary markets for financial securities are organized as brokered markets with, typically, an investment bank acting as the broker.
- In a **dealer market**, a dealer acts as the market-maker buying from the sellers and then selling to prospective buyers. For example, secondary markets for bonds are organized as dealer markets.

[1] There are exceptions to this. For example, a firm may buy back its shares on the open market. A vast majority of secondary market transactions are between investors.

- In an **auction market**, buyers and sellers gather in one venue (either physical or virtual) to buy and sell through an auction mechanism. Secondary markets for equities are typically organized as continuous auction markets where buyers submit order to buy and sellers submit orders to sell. When there is a match in price of a buy and a sell order, a transaction is happens.

In a continuous auction market, such as most secondary markets for equities, there are continuously many outstanding orders to buy and sell a security. Orders to sell are known as **ask orders** and orders to buy are known as **bid orders**. The collection of all currently outstanding asks and bids is known as an **order book**. Traders can submit two main types of orders. A **market order** is an order to buy at the lowest available ask price or to sell at the highest available bid price. A **limit order** is an order to buy or sell at a specific price. The limit orders go in the order book to wait for a market order to match. The market orders match to existing limit orders in the order book.

A trader incurs transaction costs from trading. A **bid-ask spread** is the difference between the lowest ask price and the highest bid price, essentially the difference between the highest price that a buyer is willing to pay for an asset and the lowest price that a seller is willing to accept. The wider this spread, the higher the implicit costs of trading are. The trader also has to pay a fee to her broker for facilitating the trade. This fee is known as a **commission**.

Having some money left over and buying some financial securities (such as stocks or bonds) seems quite natural. However, in modern financial markets, investors can also borrow to transact. **Buying on the margin** refers to an arrangement where an investor borrows part of the total purchase price of a security. The investor's own money makes up only a part of the total purchase price, so she borrows the rest, typically from her broker. She, of course, has to pay back the borrowed amount, plus interest, back at a later date. **Short selling** is even more exotic and amounts to selling a financial security that the investor does not own. Say, the investor thinks that a stock will decline in value in the future. To benefit from this view, she could sell the stock. But she does not own any of the stock. Not a problem, she can borrow the stock through her broker, sell the stock on the

market, and buy back and return to the lender at a later date. If she turns out to be right, and the price declines, she profits from this transaction.

1.4 Portfolio management

Much of what follows focuses on portfolios rather than individual assets. An investor's **portfolio** is simply the collection of all her investments. A portfolio can contain just one asset or multiple assets. Portfolio **allocation** refers to the composition of the portfolio, what fraction of the portfolio is invested in each asset (or asset class).

There are two big decisions every investor has to make when it comes to deciding their portfolio allocation. **Asset allocation** is the decision of how much to invest in each asset class, such as equities, bonds, and money market securities. **Security selection** is the decision about which particular assets hold and in which proportions. For example, an investor may decide on an asset allocation of 70% in equities and 30% in bonds, and within the equity part she may decide to invest 5% in the GM stock and not at all in the Tesla stock.

Finally, there are two broad approaches to portfolio construction. The **top-down** approach starts from the asset allocation into broad asset classes and then selects securities in each class. Going the other way, the **bottom-up** approach focuses on selecting the best securities to hold, and the asset allocation is merely an outcome of the security selection. Most investing wisdom favors the top-down approach.

Summary of Chapter 1

- Investing is the process of committing resources today to receive some benefits in the future. In the context of this course, investing is the purchase of financial assets in order to enjoy cash flows and price appreciation in the future.

- Real assets generate the net income of the economy, whereas financial asset determine the distribution of this income.
- Securities are financial assets that are tradable and fungible. The main classes of securities are debt (or fixed income) securities and equity securities.
- Financial markets are markets for financial securities. In addition to facilitating trade, they serve many other useful purposes.
- Firms and governments issue securities in the primary market. Investors trade securities in the secondary market.
- Buying on margin and short selling allow investors to borrow either cash or securities for the purpose of trading.
- A portfolio is the collection of all an investors financial assets.

Chapter 2

Prices and returns

This chapter establishes the general concepts of return, expected return, and risk of an investment, and discusses why prices of financial assets behave the way they do.

2.1 Price

A key ingredient in any analysis of investments and investment opportunities is the **price** of an asset. Typically, the price is set on the market as a result of matching investors' willingness to buy and sell the asset. P_t denotes the price at which an investor can buy or sell a particular asset at time t.

Theoretically, the price of an asset should equal the present value of the expected cash flows generated by the asset:

$$P_t = \sum_{\tau=1}^{\infty} \frac{E\left[CF_{t+\tau}\right]}{(1+k)^{\tau}}, \tag{2.1}$$

where $E\left[CF_{t+\tau}\right]$ are the expected future cash flows and k is the appropriate discount rate. In the chapters below, we will figure out which cash flows go in the numerator and which discount rate goes in the denominator. For now, we can

treat prices as given.

The **law of one price** states that two assets that have exactly the same future cash flows should also have exactly the same price today. This makes perfect sense. For example, all shares of Apple have the same price as they are claims to exactly the same cash future flows. If the law of one price would for some reason not hold, this would create an opportunity to make instant riskless profits. If two assets have the same future cash flows but different prices, an investor could buy the cheaper asset and short the more expensive one. She would earn the price difference as profit today. In the future, her cash flows from the two positions would exactly offset and her net cash flow in all future periods would be zero. This is known as an **arbitrage**, and the investor engaged in this activity is known as an arbitrageur. A central assumption in financial economics is that the law of one price holds and there are no arbitrage opportunities.

2.2 Return

The rate of return, or simply **return**, is the relative profit on an investment. It comprises any change in value of the investment and any cash flows (interest, dividend, or other) received by the investor. Assume an investor buys a stock at time $t-1$ for price P_{t-1}. At time t, she receives a dividend D_t and sells the stock for price P_t. The investor's **profit** from holding the stock is $P_t + D_t - P_{t-1}$. It is easier, and more intuitive, to work with return, which is simply the profit scaled by the initial investment:

$$r_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}}, \tag{2.2}$$

where r_t denotes return during the period ending at time t. This can be rewritten as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} + \frac{D_t}{P_{t-1}}, \tag{2.3}$$

where the first term is known as the **capital gain rate** (often also written as $P_t/P_{t-1} - 1$) and the second term is known as the **dividend yield**. Dividend yield is always non-negative, as stocks pay either positive dividends or no dividends at

all. Stock prices, however, fluctuate and the capital gain rate may be positive or negative.

Equation (2.2) can of course also be used to calculate returns of assets other than stocks. Any cash flows the investor receives from the asset, such as coupons from a bond or rent from real estate, are captured by the D_t term. The equation can also be used in the case the investor decided not to sell the asset at the end of time t. In such a case, the capital gain rate is not realized but the return calculation appropriately accounts for the appreciation of the asset price.

The return over multiple time periods can be calculated as the product (not the sum) of single-period returns. An asset's return from period t to period $t + \tau$ is given by

$$r_{t:t+\tau} = (1 + r_t) \times (1 + r_{t+1}) \times \ldots \times (1 + r_{t+\tau}) - 1 = \prod_{i=0}^{\tau} (1 + r_{t+i}) - 1. \quad (2.4)$$

It is important to draw a distinction between two related but very different concepts: **realized** and **future** returns. Realized return is a return that has already happened. It is a matter of historical record and there is no uncertainty associated with it. Future returns are returns that have not yet been realized. Hence, there is uncertainty, or risk, associated with them. While realized returns are important for the purposes of keeping records of investment performance, here we are more interested in future returns, especially risks and expected returns. Appendix B provides a primer on random variables which may be useful background for the analysis of future returns as random variables.

2.3 Expected return and risk

As asset prices fluctuate in an unpredictable manner, the future return from holding an asset is random. An investor will not know what her future return on an investment will be. The fact that future returns are random does not mean that we know nothing about them. We can have information about the most likely

future returns and the uncertainty related to the future returns. These are called expected return and risk, respectively, and are the two most important properties of an investment.

Expected return is the expected value of the future return. It is the most likely return to be realized in the future. If the investor holds the investment for a very long (infinite) period of time, her average return will be equal the expected return. Expected return is commonly denoted by $E[r]$ or μ (Greek letter mu).

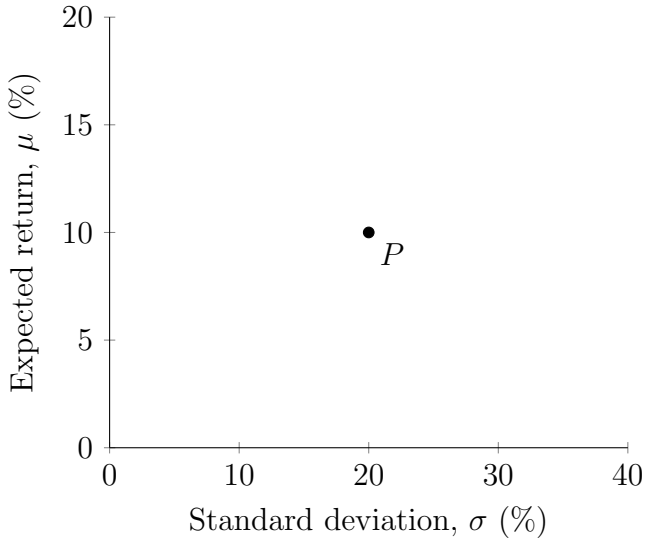
The uncertainty related to the future returns is known as **risk**. An investors may have an estimate of the expected return, but the actual return may end up being higher or lower than the expectation. Risk measures by how much the realized returns can deviate from the expected return. As is described in Appendix B, **standard deviation** is an intuitive measure of the degree of uncertainty associated with a random variable. In the finance context, the standard deviation of returns is often referred to as **volatility**. A useful rule of thumb is that if returns are **normally distributed**, 68% of the realized returns will lie within one times the volatility from the average realized return and 95% of realized returns will lie within two times the volatility from the average. A high volatility means that the potential future returns of an investment are dispersed widely around the most likely return, i.e. expected return. Such a high volatility investment hence has a high risk. The potential future returns of a low volatility, i.e. low risk, investment are clustered relatively close to the expected return. An investment that has zero volatility has no risk, as the return will for sure be equal to the expected return. Volatility is commonly denoted by $SD[r]$ or σ (Greek letter sigma).

Variance is another measure of risk. Variance is simply the square of standard deviation. Because of this, there is a one-to-one mapping between standard deviation and variance, and variance offers no additional information over standard deviation. Variance is also even less intuitive than standard deviation. However, in some applications below it is easier to work with variance than with standard deviation. Variance is commonly denoted by $Var[r]$ or σ^2.

When the investment horizon grows the standard deviation of returns, or volatility,

grows as well. However, volatility only grows by the square root of the time horizon. Hence, if the one-period volatility is σ, the T-period volatility is $\sigma \times \sqrt{T}$. For example, if the standard deviation of monthly returns is 5% then the standard deviation of annual returns is $5\% \times \sqrt{12} \approx 17.3\%$, which is only about 3.5 (not 12, as one might guess) times the monthly figure.

Figure 2.1: Risk-expected return graph.



It is often useful to think about portfolios graphically plotting their expected returns against their risks. Figure 2.1 provides an example. In the figure, portfolio P has an expected return of 10% and a volatility of 20%. Dots higher up (lower down) in the graph represent portfolios with higher (lower) expected returns. Dots more to the right (left) represent portfolios with higher (lower) risk.

2.4 Efficient market hypothesis

What moves asset prices? Why prices sometimes go up, sometimes down, and sometimes hardly move at all? As we note above in Section 2.1, the price of an

asset equals the present value of the expected cash flows generated by the asset:

$$P_t = \sum_{\tau=1}^{\infty} \frac{E\left[CF_{t+\tau}\right]}{(1+k)^{\tau}}, \tag{2.5}$$

where $E\left[CF_{t+\tau}\right]$ are the expected future cash flows and k is the appropriate discount rate. Hence, prices change when either the expected cash flows change or the discount rate changes.

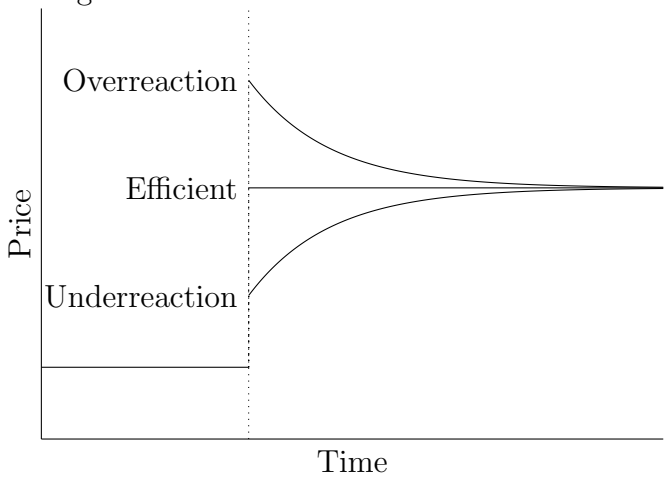
The **efficient market hypothesis** states that asset prices reflect all relevant information instantaneously. This means that when any new information relevant to the expected cash flows or the discount rate arrives, the price of the asset moves immediately to reflect this new information. This immediate price reaction is the result of a very large number of investors constantly analyzing a very large number of information sources and trading in order to take advantage of any information they may have. At the arrival of positive (negative) news, these investors rush to buy (sell) the asset pushing its price up (down) until the price on the market equals the fair value under the new piece of information. With a large number of investors competing to be the fastest to gather and process new information and trade accordingly, the price reaction to new information is instantaneous.

There are three versions of the efficient market hypothesis: weak, semi-strong, and strong. The **weak** version says that asset prices reflect all the information that can be derived from past trading data such as historical prices and trading volumes. The **semi-strong** version says that prices reflect all publicly available information. Finally, the **strong** version says that prices reflect all relevant information, even information that is not publicly available.

Figure 2.2 illustrates the price reaction to new information under efficient markets and two alternatives. The figure plots the price of an asset over time. The dotted vertical line marks the arrival of new information. Before the new information arrives, the price does not change as all inputs to Equation 2.5 remain unchanged. When the new information arrives the price moves. The efficient market hypothesis states that the price should immediately jump to its new fair value and remain there until the next piece of new information arrives. The price reaction may

also be inefficient. The two most obvious cases of inefficient price reaction are underreaction and overreaction. In the case of an **underreaction**, the price moves immediately in the right direction, but not sufficiently much. In the case of an **overreaction**, the price reaction overshoots and the price moves in the right direction by too much. In both cases, the inefficient initial price reaction is followed by a drift towards the fair value over time.

Figure 2.2: Price reaction to information.



In the real world, asset price charts do not look like any of the three lines in Figure 2.2. Prices are in constant motion up and down in patterns that can best be described as random. This, however, is perfectly consistent with the efficient markets hypothesis. New information arrives almost continuously. And the content of this new information is random by definition. Hence, also the price changes are continuous and seem random.

The efficient market hypothesis has some very strong implications for investors. If markets are efficient, there is no point in active portfolio management strategies such as trying to buy undervalued assets. As prices reflect all information, no asset is undervalued. An investor may think that she has found an undervalued asset, an asset whose market price is less than the investor's own assessment of the value. This, however, is not a result of the market price being too low. On the contrary, the investor's assessment of the value is is too high due to too optimistic cash flow expectations or a too low discount rate. If markets are efficient, the

expected return of any asset, investment, or investment strategy depends only on its riskiness, not on any advantages in information processing.[2]

A key building block underlying the efficient market hypothesis is that investors on aggregate are **rational**. This means that they process information correctly and make correct decisions given the information. In the world of the efficient market hypothesis, individual investors can make mistakes in processing information but these mistakes cancel out when all the investors' actions are aggregated in the market to form asset prices. For example, some investors may interpret new information too positively whereas other interpret it too negatively. However, in the process of trading the assets, these mistakes cancel out and the resulting asset prices reflect the new information correctly.

2.5 Behavioral finance

While the efficient market hypothesis relies on aggregate rationality of the investors, this might not be an accurate description of how real humans process information and make decisions under uncertainty. **Behavioral finance** studies systematic deviations from the assumptions of unbiased information processing and rational decision making. When processing information, people, for example, put too much weight on the most recent evidence (memory bias), overestimate the precision of their own belief (overconfidence), are too slow to update their beliefs (conservatism), and treat small samples as too representative of the population (sample size neglect). When making decisions, people react differently to the same information depending on how it is presented (framing), segregate financial decisions into various categories (mental accounting), and make decisions with a purpose of avoiding feeling regret in the future (regret avoidance). These, and many more biases documented in investor behavior, are strong enough to have significant negative impacts on investors' own financial outcomes. If many investors behave in the same biased manner, and there are not enough rational

[2]We will return to the relation between risk and expected return in Part III where we discuss asset pricing.

investors to correct their mistakes, the biases can also affect prices and returns on the aggregate level.

Summary of Chapter 2

- Return measures the relative (percentage) profit or loss on an investment.
- Expected return is the expected value of a future return.
- Risk is the degree of uncertainty related to a future return. Risk is often measured by the standard deviation of returns, also known as volatility.
- The price of an asset equals the present value of the future cash flows generated by the asset. Prices change with the arrival of new information about the cash flows or the discount rate. As new information is random, changes in prices of financial assets are also random.
- Efficient markets hypothesis states that prices reflect relevant information instantaneously. If markets are efficient, investors cannot earn higher returns by gathering and processing information and trying to identify undervalued assets.
- Behavioral finance studies the real-world deviations from the assumption of rational information processing and decision making.

Part II

Portfolio theory

Chapter 3

Portfolios

Investors typically hold (or it would optimal for them to hold) **diversified portfolios** of risky assets. This means that they hold more than one risky asset. This is done to reduce the risk of the overall portfolio. This chapter introduces the basics of portfolios with focus on the expected returns and, especially, the risks of portfolios.

3.1 Portfolios of two assets

Let us start with a two-asset portfolio.[3] An investor holds portfolio P consisting of two risky assets: A and B. The investor invests the proportion w_A in A and the remaining proportion, $w_B = 1 - w_A$, she invests in B.

Expected return of this portfolio is given by

$$\mu_P = w_A \, \mu_A + w_B \, \mu_B, \tag{3.1}$$

where μ_A and μ_B are the expected returns of A and B, respectively. This equation

[3]Investors really should not hold portfolios of just two assets. However, this simplified setting serves as a useful starting point to learn the basics of portfolio theory. We extend the analysis to more than two assets in Section 3.2.

makes intuitive sense: fraction w_A of the portfolio is expected to earn return μ_A while fraction w_B is expected to earn μ_B. The total portfolio expected return is the just a weighted average of the expected returns of the two assets.

The risk of the portfolio is a bit more complicated, and interesting, than the expected return.[4] The variance of the portfolio returns is given by

$$\sigma_P^2 = w_A^2\,\sigma_A^2 + w_B^2\,\sigma_B^2 + 2\,w_A\,w_B\,\sigma_A\,\sigma_B\,\rho_{A,B}. \tag{3.2}$$

The standard deviation of the portfolio returns, or the portfolio volatility, is given simply by taking a square root of the variance,

$$\sigma_P = \sqrt{w_A^2\,\sigma_A^2 + w_B^2\,\sigma_B^2 + 2\,w_A\,w_B\,\sigma_A\,\sigma_B\,\rho_{A,B}}. \tag{3.3}$$

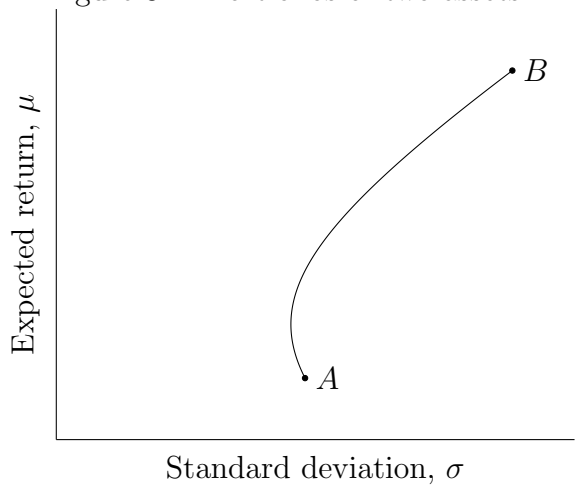
In Equations 3.2 and 3.3, σ_A and σ_B are the volatilities of A and B, respectively. But the most interesting term in these equations is the **correlation** between the returns of A and B, denoted by $\rho_{A,B}$ (Greek letter rho). The lower the return correlation between the two assets, the lower the portfolio volatility. Correlation is always between -1 and 1 and measures the degree to which the two assets move in relation to each other. A perfect positive correlation means that the correlation coefficient is exactly 1. This implies that as one asset moves, either up or down, the other stock moves in lockstep, in the same direction. A perfect negative correlation (-1) means that two assets always move in opposite directions, while a zero correlation implies no relation at all. Equation 3.3 implies that an investor can reduce the volatility, or riskiness, of her portfolio by investing in assets that have a low correlation with each other, for example stocks in unrelated industries. Reducing portfolio risk by investing in less-than-perfectly correlated asset is known as **diversification**.

Figure 3.1 provides and illustration of diversification. There are two assets, A and B, with some expected returns and volatilities. The line connecting the assets plots all the possible portfolios made out of the two assets, assuming that the correlation

[4] Appendix B discusses the calculation of variance and standard deviation of a sum of two random variables.

between the two assets is not perfect (positive or negative). One might intuitively expect that the line representing the possible portfolios is straight. That, however, is not the case. Rather, the line depicting the possible portfolios curves to the left, towards lower risk. This curving is due to the less-than-perfect correlation between the asset returns.

Figure 3.1: Portfolios of two assets.



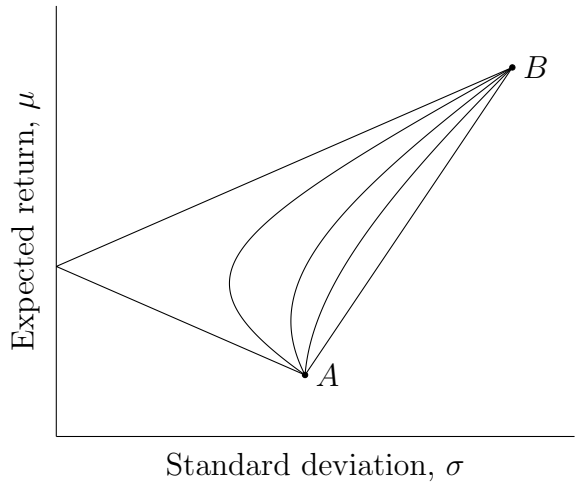
Think about this situation from the perspective of an investor who currently has all her wealth invested in asset A. She could sell some of her holdings in A and buy some of asset B instead. This would move her portfolio along the line toward asset B. It is possible for her to construct a portfolio that has a higher expected return and a lower volatility than her current portfolio. Who would not want that? This is exactly the point of diversification; by holding multiple assets, an investor can construct a portfolio that is better in terms of reward-to-risk ratio than any individual asset on its own.[5]

Figure 3.2 extends Figure 3.1 by plotting the possible portfolios made out of assets A and B, with different levels of correlation between the asset returns. Let us inspect the different lines in the graph from right to left. First, the straight line running from A to B represents the possible portfolios when the returns of two assets are perfectly correlated ($\rho_{A,B} = 1$). In this case there is no curve to the line

[5]We will return to the construction of optimal portfolios in Chapter 4.

and no benefits of diversification. This is because the perfect correlation means that returns of the two assets will always move in lockstep. Second from the right, the slightly curvy line represents the possible portfolios when the return correlation between the two assets is 0.6. The third line is identical to that in Figure 3.1 and represents the possible portfolios when the correlation is 0.2. On the fourth line, the correlation is -0.4. Finally, the fifth line, the combination of two straight lines, represents the situation where the asset returns are perfectly negatively correlated ($\rho_{A,B} = -1$). This figure shows quite simply that the smaller the correlation between the two assets the more effective diversification is. This is also easy to see from Equations 3.2 and 3.3 where plugging in a smaller value for $\rho_{A,B}$ lowers the portfolio variance and volatility.

Figure 3.2: Portfolios of two assets with different correlations.



Besides correlation, another way of measuring the comovement between asset returns is **covariance**. Covariance, commonly denoted by $Cov\,[r_A, r_B]$ or $\sigma_{A,B}$, is simply the product of the volatilities of the two assets and the correlation between the assets,

$$\sigma_{A,B} = \sigma_A\,\sigma_B\,\rho_{A,B}. \tag{3.4}$$

Covariance is not bounded like correlation and is thus less intuitive. However, it offers a useful, and more compact, way of expressing the product of volatilities and correlation that appears often in the context of portfolio theory.

3.2 Portfolios of many assets

The results above results extend easily for portfolios of more than two assets. Let us assume a portfolio made out of N assets, numbered from 1 to N. The expected return of this N-asset portfolio is given by

$$\mu_P = \sum_{i=1}^{N} w_i \, \mu_i, \tag{3.5}$$

where w_i is the portfolio weight of asset i and μ_i is the expected return of asset i.

The variance of the N-asset portfolio is given by

$$\sigma_P^2 = \sum_{i=1}^{N} w_i^2 \, \sigma_i^2 + \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} 2 \, w_i \, w_j \, \sigma_i \, \sigma_j \, \rho_{i,j}, \tag{3.6}$$

where σ_i^2 is the variance of asset i and $\rho_{i,j}$ is the correlation between the returns of asset i and asset j.

The variance formula in Equation 3.6 may look complicated and messy but it is just a generalization of the 2-asset portfolio variance formula in Equation 3.2.[6] There is a variance term for each asset i ($w_i^2 \, \sigma_i^2$) and a covariance term for each pair of assets i and j ($2 \, w_i \, w_j \, \sigma_i \, \sigma_j \, \rho_{i,j}$). The volatility of the N-asset portfolio is then simply the square root of the variance.

One way to visualize the portfolio variance is to think of an $N \times N$ matrix like in Table 3.1. Here, each line represents an asset from 1 to N and each column represents an asset from 1 to N. In the elements of the matrix are the variance and covariance terms associated with the assets represented by the corresponding row and column. The total portfolio variance is just the sum of all the element of the matrix. In the diagonal elements of the matrix are all the variances and in the off-diagonal elements are all the covariances.

Finally, it is worth noting that when the number of assets grows larger, the number

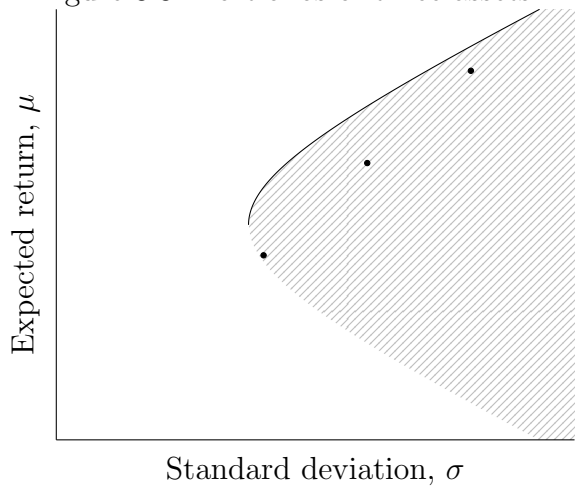
[6] To be exact, Equation 3.2 is a special case of Equation 3.6 where $N = 2$.

Table 3.1: N-asset portfolio variance.

	Asset 1	Asset 2	Asset 3	$\cdots$	Asset N
Asset 1	$w_1\, w_1\, \sigma_{1,1}$	$w_1\, w_2\, \sigma_{1,2}$	$w_1\, w_3\, \sigma_{1,3}$	$\cdots$	$w_1\, w_N\, \sigma_{1,N}$
Asset 2	$w_2\, w_1\, \sigma_{2,1}$	$w_2\, w_2\, \sigma_{2,2}$	$w_2\, w_3\, \sigma_{2,3}$	$\cdots$	$w_2\, w_N\, \sigma_{2,N}$
Asset 3	$w_3\, w_1\, \sigma_{3,1}$	$w_3\, w_2\, \sigma_{3,2}$	$w_3\, w_3\, \sigma_{3,3}$	$\cdots$	$w_3\, w_N\, \sigma_{3,N}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
Asset N	$w_N\, w_1\, \sigma_{N,1}$	$w_N\, w_2\, \sigma_{N,2}$	$w_N\, w_3\, \sigma_{N,3}$	$\cdots$	$w_N\, w_N\, \sigma_{N,N}$

of parameters that need to be estimated in order to calculate the portfolio variance grows even faster. To estimate the variance of a portfolio with N assets, one needs N variances (one for each asset) and $\frac{N(N-1)}{2}$ correlations. For example, with just two assets a total of three parameters (two variances and one correlation) are needed, whereas with 100 assets a one needs a total of 5,050 parameters (100 variances and 4,950 pairwise correlations).

Figure 3.3: Portfolios of three assets.



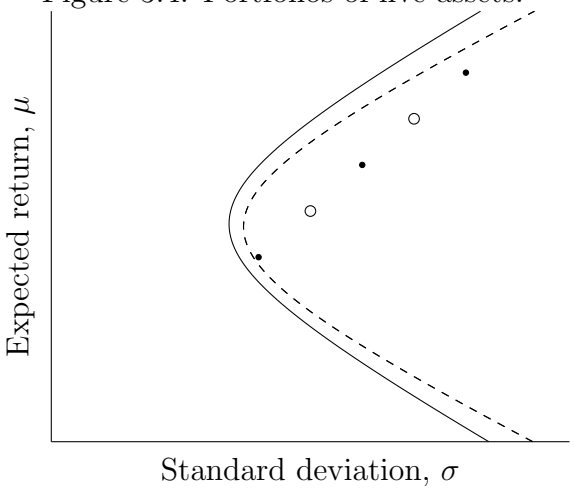
Plotting the possible portfolios of many assets is a bit more complicated than in the case of just two assets above. Rather than a line in the risk-expected return graph, the possible portfolios now form an area. Figure 3.3 illustrates the case with three assets. The assets are represented by the three dots. Using the formulas in Equations 3.5 and 3.6, one can calculate the expected returns and volatilities of all possible portfolios made out of these three assets by varying the weights of the assets. There are infinitely many such portfolios and together they form the

shaded area of the graph. Hence, an investor could pick any point in the shaded area and construct a portfolio whose expected return and volatility matches that particular point.

Many of the portfolios are **inefficient**. For many portfolios, actually for all the portfolios in the inner region of the shaded area, there exists another possible portfolio that has the same risk but a higher expected return. No investor should ever invest in such inefficient portfolios. Hence, we are mainly interested in the upper part of the border of the shaded area, plotted by the solid line. This line represents all the **efficient** portfolios that have the highest expected return for a given level of risk. Hence, it is known as the **efficient frontier**. A rational investor will only invest in an efficient portfolio. It is important to note that none of the three original assets are efficient. They all lie far below the efficient frontier. This is, again, diversification at work. By diversifying across multiple assets, investors can construct portfolios that offer better return-to-risk ratios than any of the assets individually.

The portfolio with lowest possible risk is known as the **global minimum variance portfolio**. This portfolio lies at the leftmost tip of the shaded area in the figure and is the starting point of the efficient frontier.

Figure 3.4: Portfolios of five assets.



Adding more assets to the mix naturally affects the set of possible portfolios.

Figure 3.4 presents the result of adding two more assets, marked by circles, to the three-asset example of Figure 3.3. The dashed line presents the frontier of the area of possible portfolios with just the three assets and the solid line plots the frontier with five assets. It is clear from the figure that adding the two additional assets moves the frontier to the left. This means that having access to more assets improves diversification as investors can construct even more efficient portfolios out of five assets than they could do out of three assets. It is important to note that the two additional assets are similar to the original three in terms of their expected returns and volatilities. Hence, the improvement in diversification here is not due to the new assets having particularly high expected return or low risk. The improvement is due to the less-than-perfect correlation between the returns of the assets.

3.3 Limits of diversification

The result in Figure 3.4 raises the question that what happens if the number of assets is increased to very large, or infinite. Will all the risk be diversified away such that the global minimum variance portfolio has a zero variance? To see the answer, consider a simple case with N assets, all having the same expected return μ and the same volatility σ. Also all pairs of assets have the same correlation ρ. Let us construct a simple equal-weighted portfolio of the N assets. An equal-weighted portfolio invests the same weight in each asset. With N assets, the portfolio invests $\frac{1}{N}$ in each asset. Applying the formula in Equation 3.6 and plugging in the volatilities and correlations, the variance of the equal-weighted portfolio is

$$\sigma_P^2 = \sum_{i=1}^{N} \frac{1}{N^2}\sigma^2 + \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} 2\frac{1}{N}\frac{1}{N}\sigma\,\sigma\,\rho. \tag{3.7}$$

Noting that the summations are just repeating the same terms over and over again, the portfolio variance can be written as

$$\sigma_P^2 = \frac{1}{N}\sigma^2 + \frac{N-1}{N}\sigma^2\,\rho. \tag{3.8}$$

Figure 3.5: Equal-weighted portfolios.

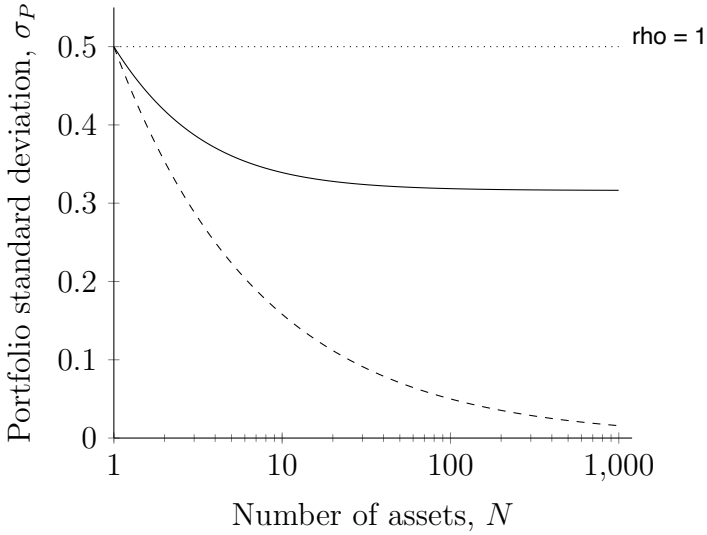


Figure 3.5 plots the standard deviation of the equal-weighted portfolio returns as a function of the number of assets N, calculated using Equation 3.8.[7] The solid line depicts the case when the volatility of each individual asset is 0.5 and the return correlation between all asset pairs is 0.4. In this case, increasing the number of assets initially lowers the portfolio volatility. However, the marginal effect of increasing N becomes smaller and smaller as N grows. Adding more assets beyond 100 has practically no impact on the portfolio volatility. The dashed line shows what happens if the pairwise correlation between the assets is zero. Now, increasing the number of assets lowers the portfolio volatility until, eventually, the portfolio becomes practically riskless when the number of assets becomes really large. Finally, the dotted line depicts the case when the assets are perfectly correlated ($\rho = 1$) and diversification does not lower the portfolio volatility at all. This is to be expected as diversification requires less-than-perfect correlation to work.

Why does the solid line in Figure 3.5 not go to zero? Letting N in Equation 3.8 grow infinitely large yields the limit of the portfolio variance as

$$\lim_{N \to \infty} \sigma_P^2 = \sigma^2 \rho. \tag{3.9}$$

[7]Note that the horizontal axis is on a logarithmic scale.

As long as the pairwise correlation between the assets is positive, the variance of an infinitely well-diversified portfolio will also be positive. Hence, while diversification can eliminate some risk, it cannot eliminate all risk. If the correlation is zero, then the limit of the portfolio variance is also zero, as seen in the dashed line in Figure 3.5.

In the real world, the returns of different assets are, on average, positively correlated. This means that there is a limit on how much investors can reduce their portfolio risk by diversification. This distinction between diversifiable and non-diversifiable risk turns out to be very important for the determination of expected returns. This will be the topic of Chapter 5.

Summary of Chapter 3

- Correlation measures the degree of comovement between the returns of two assets. Numerically, correlation is always between -1 and 1.
- Low correlation between asset returns results in diversification benefits. An investor can improve the expected return-to-risk ratio of her portfolio by investing in assets that have a low correlation with the assets she already holds.
- Portfolios that have the highest possible expected return for a given level of volatility are called efficient portfolios. Efficient frontier is the collection of all the efficient portfolios.
- There are limits of diversification if assets, on average, are positively correlated. In such a case, not all risk can be eliminated by diversifying and some risk always remains even in infinitely diversified portfolios.

Chapter 4

Portfolio optimization

This chapter works through the process of finding an optimal portfolio from the menu of infinitely many possible portfolios. The end result is surprisingly simple single portfolio of risky assets that is optimal to all investors.

4.1 Utility

To find an optimal portfolio, we first have to know what aspects of the portfolio an investor cares about. For simplicity, let us assume that an investor only cares about the expected return and the riskiness of her portfolio.[8] Given two equally risky mutually exclusive portfolios, the investor would prefer the one with higher expected return. Similarly, she would prefer the portfolio with lower risk when choosing between two portfolio with the same expected returns.

But what if the investor can choose between two portfolios where one has a higher expected return and a higher risk than the other? We need a way to map the two dimensions of a portfolio, expected return and risk, to a single value in order to compare the two portfolios. Such single value is known as **utility**. We can think

[8]This assumption ignores any non-monetary motives for investing, such as investing in a sports club to support the team rather than to earn financial returns.

of utility as the investor's degree of happiness with the portfolio. The higher the utility the happier the investor is with the portfolio. An optimal portfolio is one that gives her the highest utility. A **utility function** provides the functional form of calculating the utility a particular investor experiences from holding a particular portfolio. Let us assume that an investor's utility from holding a portfolio is given by

$$U = \mu - \frac{1}{2} A \sigma^2, \tag{4.1}$$

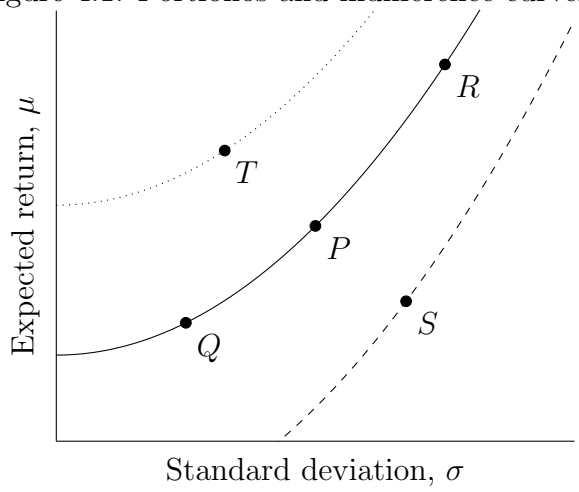
where μ is the expected return of the portfolio, σ^2 is the variance of the portfolio returns, and A is the investor's coefficient of **risk aversion**.

If an investor's coefficient of risk aversion is positive ($A > 0$), she is said to be **risk-averse**. This means that she dislikes risk and would willing to trade off some of the expected return in order to reduce risk. Most investors are assumed to be risk-averse. The higher the coefficient of risk aversion, the more the investor dislikes risk, and the more expected return they are willing to give up to decrease portfolio risk by a small amount.

If an investor's coefficient of risk aversion is zero ($A = 0$), she is said to be **risk-neutral**. A risk-neutral investor does not care about the riskiness of her portfolio, only about its expected return. If an investor's coefficient of risk aversion is negative ($A < 0$), she is said to be **risk-loving**. Such an investor wants her portfolio to have more, rather than less risk. While risk-loving behavior among investors seems odd, it can explain why people play the lottery even though it has negative expected return and high risk.

One way to represent the utility function graphically are **indifference curves** that plot all the portfolios that give the investor the same amount of utility. Figure 4.1 illustrates. It plots five portfolios (dots) and three indifference curves (lines). Assume that the investor is comparing portfolio P to the other portfolios. The solid line shows the investor's indifference curve associated with portfolio P. All dots along this line, such as portfolios Q and R, represent portfolios that give the investor the same level of utility as portfolio P. If the investor would have to choose one of these portfolios, she would like them all equally much. Another way to put is that she is **indifferent** between portfolios P, Q, and R.

Figure 4.1: Portfolios and indifference curves.



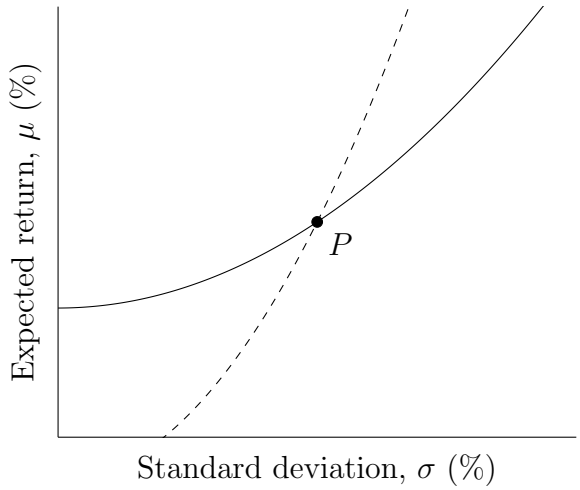
Any point below the solid line, such as portfolio S, is a portfolio that gives the investor less utility than portfolio P. Any point above the line, such as portfolio T is a portfolio that gives the investor more utility than portfolio P. The investor would prefer P over S, and she would prefer T over P. Portfolios S and T also have indifference curves associated with them, plotted in the figure by the dashed and the dotted line, respectively.

The slope of the indifference curve is determined by the investor's coefficient of risk aversion, A in Equation 4.1. As illustrated by Figure 4.2, a higher coefficient of risk aversion results in steeper indifference curves. The solid line presents the indifference curve of an investor with a relatively low coefficient of risk aversion whereas the dashed line is that of an investor with a higher risk aversion.

4.2 Portfolio optimization with one risky asset

The goal of portfolio theory is to eventually construct an optimal portfolio for an investor out of multiple assets. For simplicity, let us start with a case where the investor is constructing a portfolio out of just two assets: a risky asset P and a risk-free asset.

Figure 4.2: Risk aversion and indifference curves.



A **risk-free asset** is a special asset that offers its holder a certain return of r_f and has a volatility of zero. One can think of this as a bank account. Investing a positive amount in the risk-free asset is equivalent to depositing money in the bank. Investing a negative amount in the risk-free asset (i.e. shorting it) is equivalent to borrowing money from the bank.

Risk-free asset means that the that asset has volatitlity zero

Now, the investor invests fraction w of her wealth in the risky asset P which has expected return μ_P and volatility σ_P. The remainder of her wealth, fraction $1-w$, she invests in the risk-free asset which, as mentioned above, has expected return r_f and volatility 0. This results in a portfolio that can be denoted by C.

Plugging this information into Equation 3.1 gives the expected return of portfolio C as

$$\mu_C = w\,\mu_P + (1-w)\,r_f. \tag{4.2}$$

This is quite intuitive. A fraction w of the portfolio is expected to earn a return of μ and a fraction $1-w$ earns r_f. It turns out to be useful to rearrange the terms such that risky asset weight w appears only once and write the portfolio expected return as

$$\mu_C = r_f + w(\mu_P - r_f). \tag{4.3}$$

The difference between expected return of a risky asset and the risk-free rate of

return, $\mu_P - r_f$, is known as a **risk premium**. It measures how much more expected return a risky asset offers as compensation for risk relative to the risk-free alternative. It makes intuitive sense to focus on the risk premium rather than the expected return of a risky asset when thinking of compensation for risk. For example, a risky asset offering a 6% expected return looks very differently attractive if the risk-free rate of return is 1% than what it does if the risk-free rate is 5%. The expected return of portfolio C in Equation 4.3 is equal to the risk free rate of return plus the weight invested in the risky asset times the risk premium of the risky asset.

The variance of portfolio C can be calculated by plugging in the necessary information into the formula in Equation 3.2:

$$\sigma_C^2 = w^2 \, \sigma_P^2. \tag{4.4}$$

Note that since the variance of the risk-free asset is zero, the last two terms of the variance formula in Equation 3.2 are zero and can be dropped. Taking a square root of the variance gives the volatility of portfolio C as

$$\sigma_C = w \, \sigma_P. \tag{4.5}$$

This is also quite intuitive. Only a fraction w of the portfolio is invested in the risky asset whose volatility is σ_P.

The investor's possible portfolios form a straight line in the risk-expected return graph. But why? To see this, rearrange Equation 4.5 as

$$w = \frac{\sigma_C}{\sigma_P} \tag{4.6}$$

and plug this into Equation 4.3 to get

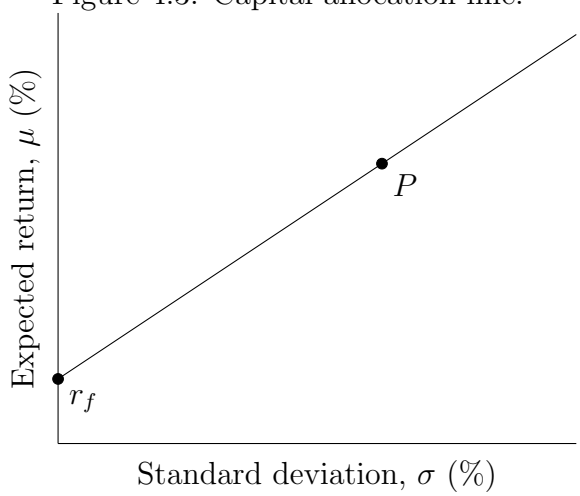
$$\mu_C = r_f + \sigma_C \frac{\mu_P - r_f}{\sigma_P}. \tag{4.7}$$

This shows that the expected return of portfolio C (μ_C) is linear in its volatility (σ_C). It is important to note that this linear relation between a portfolio's expected

return and volatility is not a general result that would apply to all portfolios of two assets. Section 3.1 shows that typically the relation is non-linear and depends on the correlation between the returns of the two assets. Here, the linearity arises because one of the assets is risk-free.

The line in the risk-expected return graph showing the possible portfolios made out of a risky and a risk-free asset is known as the **capital allocation line**. Figure 4.3 provides an illustration. If the investor invests all her wealth in the risky asset P ($w = 1$), portfolio C will naturally have the same expected return and volatility as asset P. If the investor invests all her wealth in the risk-free ($w = 0$), portfolio C will naturally have expected return of r_f and volatility zero. If the investor invests a part of her wealth in the risky asset and a part in the risk-free asset ($0 < w < 1$), portfolio C lies somewhere on the straight line between r_f and P. Finally, if the investor borrows money in the risk-free asset and invests in risky asset P ($w > 1$), portfolio C lies somewhere on the line up and right of P.

Figure 4.3: Capital allocation line.



The slope of the capital allocation line,

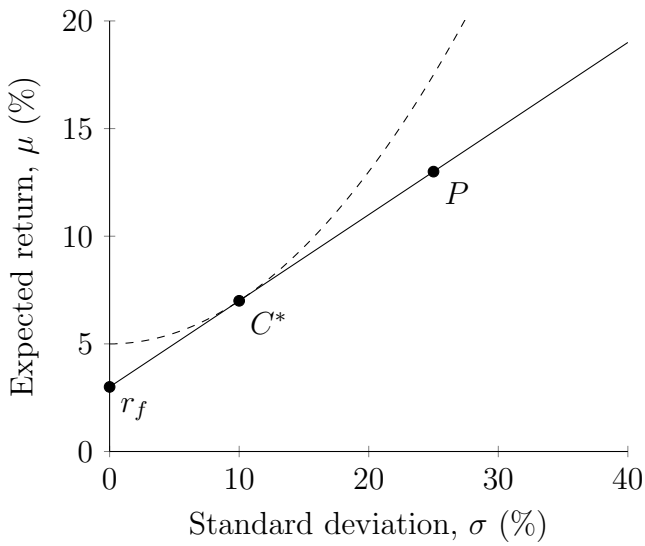
$$S_P = \frac{\mu_P - r_f}{\sigma_P}, \tag{4.8}$$

turns out to be a very useful measure of the risk-adjusted expected return of the risky asset P. It measures the ratio of the risk premium of P to the volatility of

P. This measure is commonly referred to as the **Sharpe ratio** of asset P.

How does the investor then choose the portfolio that is best for her among all the possibilities along the capital allocation line? Let us find the solution first graphically and then analytically. Figure 4.4 presents the process of finding the optimal portfolio, C^*, graphically. The investor must choose one portfolio from the solid line. She does this by finding the portfolio that gives her the highest utility. This portfolio is found at the point where the highest possible indifference curve just touches the capital allocation line. Any other point along the capital allocation line would be on a lower indifference curve and would give the investor a lower utility than C^* does. Hence, C^* must be the optimal portfolio that gives the investor the highest possible utility.

Figure 4.4: Optimal portfolio.



The optimal portfolio composition can also be solved for analytically. Remember that the investor invests w in the risky asset P, $1-w$ in the risk-free asset giving her a portfolio with expected return $\mu_C = r_f + w(\mu_P - r_f)$ and volatility $\sigma_C = w\,\sigma_P$. She now has to decide which w gives her the highest utility. To do this, we can plug the expected return and volatility into the utility function in Equation 4.1:

$$U_C = r_f + w(\mu_P - r_f) - \frac{1}{2}A\,(w\,\sigma_P)^2\,. \tag{4.9}$$

Next, we calculate the derivative of the utility with respect to w,

$$\frac{\partial U_C}{\partial w} = \mu_P - r_f - A\, w\, \sigma_P^2, \tag{4.10}$$

and find the optimal w^* by setting the derivative equal to zero:

$$\mu_P - r_f - A\, w^*\, \sigma_P^2 = 0 \Rightarrow w^* = \frac{1}{A} \frac{\mu_P - r_f}{\sigma_P^2}. \tag{4.11}$$

This solution implies that the investor invests more of her wealth in the risky asset P when i) she is less risk-averse (her A is smaller), ii) the risky asset offers a higher risk premium ($\mu_P - r_f$ is larger), and iii) the risky asset is less risky (σ_P is smaller). All these make intuitive sense.

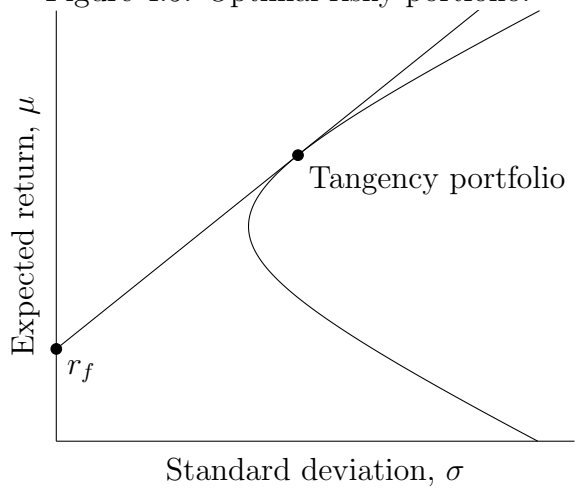
4.3 Portfolio optimization with many risky asset

Above we analyze the situation where the investor allocates between the risk-free asset and only one risky asset. But what about the more realistic case where she has access to multiple risky assets in addition to the risk-free asset? How should she allocate her portfolio? The easiest way to solve this problem is to break it in two separate steps. In the first step we find the optimal portfolio consisting of only the risky assets. In the second step we find the optimal allocation between this risky portfolio and the risk-free asset.

Figure 4.5 illustrates the first step. Like in Figure 3.3, there are many risky assets (not displayed) from which the investor could form infinitely many different portfolios. The parabola plots the frontier of the set of possible portfolios (equivalent to the shaded area in Figure 3.3). We only need to concern ourselves with the frontier portfolios as all the portfolios within the set are dominated by some frontier portfolios. There is also a risk-free asset.

The investor's first problem is to pick the best possibly risky portfolio (a portfolio made out of risky assets only) to combine with the risk-free asset. Remember, the portfolios made out of the risk-free asset and any risky portfolio form a straight

Figure 4.5: Optimal risky portfolio.



line in the volatility-expected return plot. Hence, whatever risky portfolio the investor picks to combine with the risk-free asset, will result in a set of possible portfolios represented by a straight line running from the risk-free asset through the investor's chosen portfolio. As the investor wishes to have the highest total portfolio expected return for any level of portfolio volatility, she wants to pick a risky portfolio that results in a capital allocation line with the highest possible slope. This particular risky asset portfolio is known as the **tangency portfolio** because the straight line drawn from the risk-free asset through it just touches, or tangents, the efficient frontier. Another way of saying the tangency portfolio results in the highest slope of the capital allocation line is to say that the tangency portfolio has the highest Sharpe ratio among all the possible risky asset portfolios.

Having now found the best possible risky portfolio, the investor decides the optimal allocation between the risk-free asset and the tangency portfolio. This however, is equivalent to allocating between the risk-free asset and one risky asset discussed above. The investor chooses the total portfolio that maximizes her utility, just like above in Section 4.2.

It is noteworthy that the first step, finding the tangency portfolio, does not involve the investor's preferences at all. Whatever her risk aversion happens to be, the tangency portfolio will always be the same. This is because the efficient frontier

and the location of the tangency portfolio only depends on the expected returns and risks of the assets available. All investors who agree on the expected returns, volatilities, and correlations will invest in the same tangency portfolio. In the second step the investor finds the optimal mix between the risk-free asset and the tangency portfolio. This decision depends on her risk aversion. More (less) risk-averse investor invests less (more) in the tangency portfolio.

Summary of Chapter 4

- Utility measures how satisfied an investor is with a portfolio that has a given expected return and a given level of risk. Higher utility is better.
- Risk aversion measures an investor's dislike for risk. A coefficient of risk aversion is needed in a utility function to relate expected returns and risks.
- An indifference curve plots all the portfolios that give the investor the same level of utility.
- Portfolio optimization problem can be broken into two steps. In the first step the investor finds the risky asset portfolio with the highest Sharpe ratio, known as the tangency portfolio. In the second step, she allocates her wealth between the risk-free asset and the tangency portfolio to maximize her utility.

Part III

Asset pricing

Chapter 5

Asset pricing theory

This chapter discusses asset pricing, the process of determining expected returns of assets from the theoretical point-of-view.

5.1 Asset pricing

So far, our perspective has been that of a very small investor who takes asset expected returns, volatilities, and correlations as given. However, from the perspective of the market as a whole, expected returns are really not exogenously given. Rather, expected returns depend on the investors' decisions to buy and sell assets.

To illustrate the mechanism, remember that the price of an asset must equal the present value of the expected future cash flows generated by the asset, appropriately discounted:

$$P = \sum_{t=1}^{\infty} \frac{E\left(CF_t\right)}{(1+\mu)^t}. \tag{5.1}$$

In this equation, the discount rate equals the expected return of the asset μ. Now assume that the asset becomes more popular among investors without any changes to the expected cash flows. As more investors flock to buy the asset, its

price increases. As the pricing identity must hold always, an increase in the price must be matched by a decrease in the expected return. Hence, investors' decision to buy the asset result in it having a lower expected return.

Asset pricing studies the determination of expected returns as a function of investors' demand for assets. The typical chain of logic goes as follows: i) investors dislike risk, ii) investors do not wish to hold a risky asset, iii) demand for the risky asset is low, iv) the price of the risky asset is low, and v) the expected return of the risky asset is high. Conversely, investors like safe assets, which results in safe assets having high prices and low expected returns.

As an investor can earn the risk-free rate of return r_f with no risk, it makes sense to focus on the risk premium, the expected return of a risky asset over the risk free rate: $\mu - r_f$. In standard asset pricing models, risk premium of an asset is linearly related to the risk of the asset. But what risk?

5.2 Systematic and idiosyncratic risk

Figure 3.5 in Section 3.3 plots the development of portfolio volatility as more assets are added to and equal-weighted portfolio. If the correlation between the asset returns is greater than zero, some risk always remains no matter how many assets are added to the portfolio. This means that only a part of the risk of a portfolio can be diversified whereas some part of the risk is non-diversifiable.

This gives us a decomposition the risk of any portfolio (composed of one or many assets) into two distinct components: **diversifiable** and **non-diversifiable** risk. Non-diversifiable risk, also known as **systematic** risk, is that part of the total risk that cannot be diversified away by adding more assets to the portfolio. This part of the risk is associated with the variation in returns that is common to all assets. The diversifiable risk, also known as **idiosyncratic** risk, is the part of the risk that can be diversified away. It is related to the asset-specific variation in returns. For example, returns of all stocks react to news about macroeconomic conditions, such as inflation. This is systematic risk that cannot be diversified. Conversely, the

rejection of a key patent application is very bad news for a small pharmaceutical company and its stock is likely to fall in value as a result. However, other companies are unaffected by this news, and the risk associated with the patent application is firm-specific and diversifiable.

What part of the total risk affects risk premiums? It is easiest to intuit this by thinking of the risk premium as compensation for investors for bearing the risk related to holding an asset. The investors can always diversify away all idiosyncratic risk. Hence, they do not need to be compensated for bearing it. However, they cannot, by definition, diversify away the systematic risk. As investors always have to bear the systematic risk, they need to receive a compensation for it in the form of a risk premium. Hence, risk premium depends on the systematic risk, not on the idiosyncratic risk.

Now that we have established that the risk premium should only depend on the systematic risk, two big questions remain. How do we measure the systematic risk of an asset? And, how are systematic risk and risk premium related? To answer these questions, we need an **asset pricing model**. An asset pricing model is a simplified portrayal of how investors perceive risk and how the financial markets price assets and determine expected returns. An asset pricing model provides a methodology for quantifying the systematic risk and translating it into a risk premium. The most commonly used asset pricing model is called (somewhat confusingly) the capital asset pricing model.

5.3 Capital asset pricing model

The capital asset pricing model, or simply the **CAPM**, is a bedrock of asset pricing. Introduced in the early 1960's, it is still widely used in various applications and it has spurred a large literature developing more complex models to price assets. A key advantage of the CAPM is its simplicity. It yields a simple and intuitive measure of systematic risk and a simple formula for connecting the systematic risk to a risk premium. Basically, the CAPM provides the answer to the

following question: What what would the expected returns be if all investors were rational mean-variance optimizers and behaved as we prescribe above in Section 4.3?

Let us start with the main assumptions underlying the CAPM:

- There are two kinds of **assets**: a risk-free asset and a number of risky assets.
- All investors can buy and sell all assets at **market prices**. All investors can lend and borrow at the risk-free rate. Investors incur no transaction costs or taxes. There are no other frictions.
- All investors hold **efficient** portfolios. They hold portfolios that yield the highest return for a given level of volatility. Some investors hold portfolios with higher volatility, some hold portfolios with lower volatility, depending on their individual preferences.
- All investors have **identical expectations** regarding the expected returns, volatilities, and correlations of all securities. The investors do not disagree.

Solving for the market equilibrium (the investors' portfolios and the prices of the securities) under these assumptions yields three main results. First, and least surprisingly given what we know from Section 4.3, all investors hold a combination of the risk-free asset and the tangency portfolio, or **market portfolio**, of the risky asset. While the investors' total portfolios differ—as they hold different amounts of the risk-free asset depending on their preferences—they all hold the same portfolio of risky assets.[9] In practical applications, the market portfolio is typically proxied with a value-weighted stock market index (such as the S&P 500 index in the United States).

Second, and most importantly, the CAPM produces a very simple formula for the

[9]This is not the case in the real world, as investors hold very different portfolios of risky assets. This is sometimes used as an argument against the CAPM and in favor of more complicated asset pricing models.

risk premium of an asset. In the CAPM, the risk premium of asset i is given by

$$\mu_i - r_f = \beta_i \left(\mu_M - r_f\right). \tag{5.2}$$

The risk premium equals the amount of systematic risk ,β_i, times the compensation for bearing one unit of systematic risk price of risk, $\mu_M - r_f$. μ_M stands for the expected return of the market portfolio of risky assets. $\mu_M - r_f$ is known as the **market risk premium**.

An asset's **beta**, β_i, is the measure of its systematic, non-diversifiable, risk. Beta is defined as

$$\beta_i = \frac{\sigma_{i,M}}{\sigma_M^2} = \rho_{i,M} \frac{\sigma_i}{\sigma_M}, \tag{5.3}$$

where $\sigma_{i,M}$ is the covariance between the returns of asset i and the market portfolio, $\rho_{i,M}$ is the correlation between the returns of asset i and the market portfolio, σ_i is the volatility of asset i, and σ_M is the volatility of the market portfolio. Since the correlation can have either sign and the volatility can be very large, beta can (theoretically) take any real number value.

Why is an asset's risk premium given by the product of its beta and the market risk premium? The CAPM assumptions imply that all investors hold the market portfolio of risky assets. It is easy to see that the contribution of asset i to the total risk premium of the market portfolio is $w_i \left(\mu_i - r_f\right)$, where w_i is the weight of asset i in the market portfolio. It also turns out that the contribution of asset i to the total variance of the market portfolio is $w_i \, \sigma_{i,M}$. Hence, the reward-to-risk ratio of asset i, from the perspective of an investor holding the market portfolio, is given by

$$\frac{w_i \left(\mu_i - r_f\right)}{w_i \, \sigma_{i,M}} = \frac{\mu_i - r_f}{\sigma_{i,M}}. \tag{5.4}$$

In an equilibrium, all assets must have the same reward-to-risk ratio. If one asset offers a higher ratio, all investors would buy that asset pushing up its price and pushing down its expected return. This would go on until the reward-to-risk ratio of the asset is in line with other assets. If one asset has a particularly low reward-to-risk ratio, investors would sell that asset causing its price to drop and its expected

return to increase, again, until the ratio is in line with other assets. Hence, all assets, including the market portfolio, must have the same reward-to-risk ratio. For the market portfolio, the relevant reward-to-risk measure is simply its risk premium relative to its variance. Since all assets have the same reward-to-risk ratio, it must hold for asset i and the market portfolio that

$$\frac{\mu_i - r_f}{\sigma_{i,M}} = \frac{\mu_M - r_f}{\sigma_M^2}. \tag{5.5}$$

We can multiply both sides of the equation by $\sigma_{i,M}$ to get

$$\mu_i - r_f = \frac{\sigma_{i,M}}{\sigma_M^2} \left(\mu_M - r_f\right). \tag{5.6}$$

So, there it is! Risk premium of asset i is given by the product of $\frac{\sigma_{i,M}}{\sigma_M^2}$, which is exactly equal to beta in Equation 5.3, and the market risk premium.

Next, let us try to understand the intuition behind Equations 5.2 and 5.3. First, think of an asset that has beta equal to zero. This asset has no systematic risk at all; all its risk is diversifiable. As such, it does not contribute at all to the total riskiness of the investors' portfolios. Hence, investors are willing to pay a relatively high price for this asset. This is equivalent to saying that the investors require a relative low expected return for the asset. Indeed, the risk premium of a zero-beta asset is zero and its expected return equals the risk-free rate of return. This makes sense, because from a well-diversified investor's perspective this asset does not carry any relevant, non-diversifiable, risk.

Next, think of an asset that has beta equal to one. This asset is as risky as the market portfolio, as the market portfolio also has a beta of one.[10] As this asset is as risky as the market portfolio, it will also have the same risk premium as the market portfolio.

An asset with beta greater than one has more systematic risk than the market portfolio. The returns of this asset are positively correlated with the market port-

[10] It is easy to see that the market portfolio beta equals one. Plug $i = M$ in Equation 5.3 to get $\beta_M = \rho_{M,M} \frac{\sigma_M}{\sigma_M} = 1$.

folio and its volatility is higher than the market volatility. This is not a particularly nice asset from the investors' perspective as it contributes a lot to the riskiness of their portfolios. Hence, the investors are only willing to pay a relatively low price for the high-beta asset, which is equivalent to saying that they demand a relatively high risk premium to hold the asset. This is clear from Equation 5.2 where the risk premium of a high-beta asset is higher than that of the market portfolio. An asset with beta between zero and one correlates positively with the market portfolio but has a lower systematic risk and, hence, a lower risk premium.

Assets with negative betas are interesting. To have a negative beta, the asset's returns need to be negatively correlated with the market portfolio returns. This means that when the investors' portfolios have a negative returns, the negative-beta asset is likely to have a positive return. Hence, the asset actually decreases the overall riskiness of the investors' portfolios. The investors like this insurance-like feature. Hence, they are willing to pay a very high price for the negative-beta asset. Equivalently, they are willing to accept a very low expected return, and a negative risk premium, to hold a negative-beta asset.

Empirically, beta can be estimated as the covariance between the asset's return and the market return divided by the market return variance. Identically, it can be estimated as the correlation of the asset's return with the market return times the ratio of the return volatilities (asset divided by market). By a lucky coincidence, beta corresponds to the slope coefficient in a regression of the asset's excess return on the market excess return. Excess return refers to the return of the asset minus the risk-free rate of return. We can denote excess return over time period t by R_t and define it as $R_t = r_t - r_f$. Thus, one additional way to estimate an asset's beta is to collect realized excess returns of the asset, the market index, and the risk-free asset and estimate the regression

$$R_{i,t} = \alpha_i + \beta_i \, R_{M,t} + e_{i,t}. \tag{5.7}$$

This regression directly provides an estimate of the beta.[11]

[11] Typically, either daily, weekly, or monthly returns are used to estimate the regression. As the risk-free rate of return varies over time, the regression needs to be estimated on excess returns,

Third, and finally, the CAPM provides a formula for the price of risk, i.e. the amount of risk premium investors earn for one unit of beta. As can be seen from Equation 5.2, increasing an asset's beta by one increases its expected return by an amount equal to the **market risk premium**, $\mu_M - r_f$, the risk premium of the market portfolio. In an equilibrium, all investors' borrowing in the risk-free asset must be matched by other investors investing in the risk free asset. This market clearing condition tells us that the market risk premium in the CAPM is equal to the variance of the market portfolio (σ_M^2) times the average degree of risk aversion of the investors ($\bar{A}$):

$$\mu_M - r_f = \bar{A}\,\sigma_M^2. \tag{5.8}$$

This, however, is not very operational as there is no way to measure the investors' average degree of risk aversion. Hence, the typical approach to estimating the market risk premium is to rely on the average realized market return, in excess of the risk-free rate, over a very long period of time, or surveys of investors' market return expectations.

5.4 Factor models

It is useful to note the connection between the CAPM and something known as **factor models**. A factor model posits that an asset's return (realized return, not the expected return) over the risk-free rate can be expressed as a sum of three components: a time-invariant constant return, a return due to exposure to a systematic factor, and an idiosyncratic (or asset-specific) return. Factor models differ in terms of which factors are used to model the systematic component of returns.

The simplest example of a factor model is the **single-index model**. In a single-index model, the systematic factor is the return of a market index. Hence, in the

i.e. the returns minus the risk-free rate of return each period. The maturity of the risk-free rate should match the frequency of the return observations. For example, if the observations are monthly returns, the risk-free rate should be the interest rate of a riskless one-month deposit or government bill.

single-index model, the excess return of asset i in time period t (R_i, t) is composed of the time-invariant component (α_i), an exposure to the market excess return $(\beta_i \, R_{M,t})$, and an idiosyncratic component $(e_{i,t})$:

$$R_{i,t} = \alpha_i + \beta_i \, R_{M,t} + e_{i,t}. \tag{5.9}$$

A single-index model (or factor models more generally) is a convenient way to model asset returns in ways that capture many stylized facts of returns in the real world. First of all, since all assets have some beta exposure to the same market index return, the asset returns, on average, are positively correlated. Just like in the CAPM, assets with (low) high betas comove more (less) strongly with the market index. The idiosyncratic returns $(e_{i,t})$ are uncorrelated across assets. This results in correlations between assets to be less than perfect. The idiosyncratic return is also assumed to have an expected value of zero. Finally, the time-invariant alpha (α_i) represents the non-zero average excess return of asset i which is not due to the beta exposure to the market index return.

Taking expectations of both sides of Equation 5.9, and noting that the expected excess return is the same thing as risk premium, yields

$$\mu_i - r_f = \alpha_i + \beta_i \, (\mu_M - r_f) \, . \tag{5.10}$$

This is almost the same as the CAPM in Equation 5.2. The only difference is that α_i appears here but not in the CAPM. This is an important insight. The CAPM really just implies that all expected returns are due to the beta exposures to the market return and all alphas are zero. We will use this insight to test the empirical validity of the CAPM in Section 6.1.

The variance of asset return in the single-index model is given by

$$\sigma_i^2 = \beta_i^2 \, \sigma_M^2 + \sigma_{e_i}^2, \tag{5.11}$$

where $\beta_i^2 \, \sigma_M^2$ is the systematic variance and $\sigma_{e_i}^2$ is the idiosyncratic variance.

Portfolios are also easy to analyze in the single-index model. A portfolio P consists

of N assets with w_i denoting the weight of asset i. The excess return of the portfolio is modeled as

$$R_{P,t} = \alpha_P + \beta_P \, R_{M,t} + e_{P,t}. \tag{5.12}$$

The portfolio's alpha, beta, and idiosyncratic return are simply given by

$$\alpha_P = \sum_{i=1}^{N} w_i \, \alpha_i \quad , \quad \beta_P = \sum_{i=1}^{N} w_i \, \beta_i \quad , \text{ and } \quad e_{P,t} = \sum_{i=1}^{N} w_i \, e_{i,t}. \tag{5.13}$$

Since the idiosyncratic returns of different assets are by definition uncorrelated, the idiosyncratic variance of the portfolio is

$$\sigma^2_{e_P} = \sum_{i=1}^{N} w_i^2 \, \sigma^2_{e_i}. \tag{5.14}$$

Summary of Chapter 5

- Asset pricing studies how asset prices and expected returns are set in a market equilibrium.
- Investors can diversify away all idiosyncratic risk and, hence, do not need to be compensated for bearing it.
- Systematic risk cannot be diversified away and investors need to receive a compensation, in terms of a risk premium, for bearing systematic risk.
- The capital asset pricing model (CAPM) provides a simple way to measure an asset's systematic risk and to translate it into a risk premium or an expected return. In the CAPM, systematic risk is quantified by beta which measures the covariance of the asset's returns with the market portfolio returns relative to the variance of the market returns.
- Factor models provide a simple way to model realistic asset returns. The CAPM is closely related to the single-index model, which a commonly used simple factor model.

Chapter 6

Asset pricing empirics

Chapter 5 above focuses on the theory of asset pricing, especially the CAPM. In this chapter, we are going to study the empirical validity of the CAPM in describing expected returns and then look into some other empirical regularities related to asset prices.

6.1 Testing the CAPM

The CAPM is a very popular model for calculating expected returns, mainly due to its simplicity and intuitive appeal. But how well does the model describe investors' return expectations? The answer is: not that well.

The CAPM, in Equation 5.2, states that the risk premium depends on beta, and only beta, and the price of risk equals the market risk premium. We can use data on asset returns to test whether these predictions are true in the real world.

To test the CAPM, we take the following steps:

1. Collect historical returns of N test assets, the market portfolio, and the risk-free rate asset. The test assets can be individual assets (like stocks or bonds) or portfolios of assets. We are going to need long time series of returns. It

is common to use decades of monthly returns. Denote by T the number of return observations (typically months) we have for each test asset. Denote the excess return of a test asset i ($i = 1, \ldots, N$) in time period t ($t = 1, \ldots, T$) by $R_{i,t}$ and the market portfolio excess return by $R_{M,t}$.

2. Estimate beta for each test asset. The betas are easiest to calculate by estimating a regression where the test asset excess return is the dependent variable and the market excess return is the independent variable:

$$R_{i,t} = \alpha_i + \beta_i \, R_{M,t} + e_{i,t}. \tag{6.1}$$

   This regression is estimated for each test asset separately. This step produces N beta estimates, one for each test asset.

3. Calculate the average average excess return for each test asset. Asset i's average excess return is

$$\bar{R}_i = \frac{1}{T} \sum_{t=1}^{T} R_{i,t}. \tag{6.2}$$

   This step produces N average returns, one for each test asset.

4. Relate the average excess returns to the betas. In the cross-section, estimate a regression where the average return is the dependent variable and the estimated beta is the independent variable:

$$\bar{R}_i = \gamma_0 + \gamma_1 \, \beta_i + u_i. \tag{6.3}$$

   There are N observations in this regressions, one for each test asset, and γ_0 and γ_1 are the parameters to be estimated.

The regression in Equation 6.3 is an empirical version of the theoretical CAPM in Equation 5.2. On the left-hand side of the theoretical equation, we have the risk premium. In the empirical equation we substitute it with the long-run average excess return as there is no way to credibly measure investors' expectation directly. Hence, the left-hand sides of the two equations are basically equivalent. On the right-hand side of the empirical equation we have the γ_0 coefficient plus the γ_1

coefficient times beta, whereas the right-hand side of the theoretical equation has beta times the market risk premium. From this we can see that the CAPM implies that γ_0 should equal zero and γ_1 should equal the average excess return of the market portfolio.

Researchers have tested this prediction over and over again and the findings are very consistent: in the data γ_0 is significantly greater than zero and γ_1 is significantly less than the average excess return of the market portfolio. This implies that investors do not seem to earn as much extra risk premium for holding higher-beta assets as the CAPM implies. A graph depicting risk premium (on the vertical axis) as a function of beta (on the horizontal axis) is commonly called the **security market line**. The empirical evidence shows that the true security market line is much flatter predicted by the CAPM. Since the difference between the theoretical prediction and the empirical reality is both statistically and economically very significant, this is evidence that the CAPM is not the true asset pricing model and does not appropriately describe how investors set return expectations and price assets.

Another way to investigate the empirical validity of the CAPM is to test whether other factors besides beta affect risk premiums. The CAPM predicts that only beta affects the risk premium. If we find empirically that also other factors affect risk premiums, this is direct evidence against the CAPM being the true asset pricing model.

We can easily augment the regression model in Equation 6.3 to include other factors. For example, we could use data on stock returns to test the CAPM and include the market capitalization of the stock as an independent variable in the cross-sectional regression.[12] Fama and French (1992)[13] do exactly this and report that the coefficient of the market capitalization is negative and statistically significant. This means that larger firms tend to have lower stock returns than smaller firms. This finding might be very intuitive and reasonable but it is in

[12]Market capitalization measures the market value of a firm's equity and is calculated simply as the share price multiplied by the numbers of shares outstanding.

[13]Eugene F. Fama and Kenneth R. French, "The Cross-Section of Expected Stock Returns," *Journal of Finance*, 1992.

direct contradiction of the CAPM predictions. The CAPM predicts that only beta matters, but the evidence suggests that firm size matters as well. This, again, is evidence that the CAPM is not the true asset pricing model.

Firm size is not the only factor research have found to affect risk premiums. Indeed, over the past three decades, researchers have discovered over 300 variables that affect risk premiums. We will review some of the most prominent ones in Section 6.2. Since none of these are supposed to affect risk premiums according to the CAPM, the empirical evidence against the CAPM being the correct asset pricing model is very strong.

Why does the CAPM fail empirically? A number of explanations have been offered but no consensus on one simple explanation has been reached. An obvious candidate for the empirical failure is that the assumptions underlying the CAPM are very strict and unrealistic. Relaxing the assumptions results in different asset pricing equations with different measures and prices of systematic risk. On a more methodological side, the empirical failure of the CAPM could also be due to the fact that the true market portfolio (the portfolio of all assets held by the investors) is not observable as it contains many non-traded assets such as human capital.

6.2 Cross-section of stock returns

Lots of the literature on empirical asset pricing focuses on explaining the cross-sectional differences in expected returns of stocks. This is likely driven by a combination of availability of data, large variation in realized returns across stocks, and broad interest in stocks as investment. The literature on the cross-section of expected stock returns tries to explain why some stocks seem to have higher expected returns than others. As mentioned above, this literature has documented over 300 different stock-level variables that seem to be related to expected returns.

Above we already saw that firm size, measured by market capitalization, is associated with expected returns: small firms tend to have higher stock returns than larger firms. This is known as the **size** effect. Fundamentally, such systematic

difference in expected returns can be due to two different reasons. First, small firms may be riskier than large firms. If that is the case, investors require a higher expected return to hold a small firm stock than they do to hold a large firm stock, which would explain the empirical observation. Alternatively, the difference may be due to systematic mispricing. Investors may systematically underestimate the value of small firms and, hence, only be willing to pay a too low price for small firm stocks. As the mispricing corrects over time, realized returns on small stocks would be higher than realized returns on large stocks. Disentangling the two explanations is very difficult, if not impossible.

In addition to size, value and momentum are two empirical phenomena that are discovered in different asset classes, different geographical regions, and different time periods. These are commonly regarded as the most prominent patterns in the cross section of stock returns.

The **value** effect refers to the tendency of stocks with low valuation ratios (commonly referred to as value stocks) to have higher returns than stocks with high valuation ratios (referred to as either growth stocks or glamour stocks). A valuation ratio in this context is typically defined as a market-based value of the firm's equity (stock price or market capitalization) relative to an accounting-based measure of the firm's value (such as book value of equity or earnings per share). The most common valuation ratio used in academic studies is the ratio of market capitalization to the book value of the firm's equity. Stocks whose market price is cheap compared to the book value, i.e. stocks that have low market-to-book ratios, have historically had higher returns than stocks whose market price is high relative to the book value. So-called value investors exploit this phenomenon buy investing in value stocks and avoiding, or even shorting, glamour stocks. This is known as value investing.

Why do value stocks have higher returns than glamour stocks? As we noted above, this could be due to either risk or mispricing. The risk explanation would hold that value stocks are somehow riskier than glamour stocks. The mispricing explanation would imply that investors somehow underprice the value stocks and/or overprice the glamour stocks. There exists empirical evidence in support of both

explanations.

The **momentum** effect refers to the tendency of stocks with high past returns (commonly referred to as winner stocks) to have higher future returns than stocks with low past returns (loser stocks). This effect works at horizons of a few months. The typical momentum strategy used in academic studies measures the past returns over the previous 12 months and future returns over the next one month. An investor employing the momentum strategy buys the winner stocks and shorts the loser stocks.

Also the momentum effect could be explained by risk or mispricing. While the long-term average returns to the momentum strategy are large positive, the strategy tends to suffer infrequent but large losses. These are known as momentum crashes and have so far occurred in 1932 and 2009. Hence, the momentum strategy is exposed to a crash risk that may explain its positive returns. The momentum effect could also be due to mispricing. For example, investors could underreact to new information which results in stock prices to drift slowly towards its efficient value as in Figure 2.2. There is empirical evidence supporting the mispricing explanation of the momentum effect.

6.3 Multifactor models

The fact that expected returns seem to depend on many factors has given rise to the use of factor models with multiple systematic factors. These are known as **multifactor models**. Just like the single-index model in Section 5.4, the multifactor models break the excess return in three components: a time-invariant constant return, a return due to exposures to systematic factors, and an idiosyncratic return. The only difference is that there are many systematic factors in multifactor models.

Probably the most commonly used multifactor model is the so-called **three-factor model** by **Fama and French** (1993).[14] As the name suggests, there are three

[14]Eugene F. Fama and Kenneth R. French, "Common Risk Factors in the Returns on Stocks

systematic factors in this model: the market factor, the size factor, and the value factor. The realized excess return under the Fama-French three-factor model is given by

$$R_{i,t} = \alpha_i + \beta_{M,i}\, R_{M,t} + \beta_{SMB,i}\, SMB_t + \beta_{HML,i}\, HML_t + e_{i,t}. \tag{6.4}$$

This is very similar to the single-index model in Equation 5.9; we have only added two new factors: SMB and HML. SMB is the size factor and is defined as the return difference between small firm stocks and large firm stocks. The name is simply the acronym for "small minus big". This factor captures the systematic patterns in return difference between small and large firms. HML is the value factor and is defined as the difference in returns between value and glamour firms. Fama and French measure value by the book-to-market ratio (rather than market-to-book as we do above). Hence, HML stands for "high minus low" and is the return of stocks with high book-to-market ratio (i.e. low market-to-book ratio) minus the return of stocks with low book-to-market ratio (i.e. high market-to book ratio).[15] This factor captures the systematic patterns in returns between value and glamour firms. The third factor is simply the market excess return ($R_{M,t}$), just like in the single-index model. Each asset has three factor exposure, or betas, one for each systematic factor.

The Fama-French three factor model is not theory-based like the CAPM. There is no theory behind the model explaining why returns should depend on these three factors. The model is more evidence-based. Fama and French show that the model explains a large part of the variation in expected returns and therefore consider it an appropriate model to describe asset returns. The model seems to work in practice even though there is no solid theory underlying why it works.

and Bonds," *Journal of Financial Economics*, 1993.

[15]The construction of the Fama-French value factor is a bit confusing considering that practitioners typically define valuation ratios as a market value divided by a book value. However, this is how they chose to construct their variable and this has become the convention in the academic literature.

Summary of Chapter 6

- The CAPM can be empirically tested relating average excess returns to betas and other factors. The CAPM implies that i) one unit increase in beta should lead to an increase in average excess return equal to the average excess return of the market portfolio, and ii) no other factor besides beta should affect average excess returns.

- Empirical evidence is strong against the CAPM being the true asset pricing model. The empirical security market line is too flat and many other factors than beta affect average returns.

- Three prominent effects in the cross-section of stock returns are size (small stocks have higher returns than large stocks), value (relatively cheap stocks have higher returns than relatively expensive stocks), and momentum (winner stocks have higher returns than loser stocks).

- Multifactor models expand the simple single-factor models by including many systematic factors. In the Fama-French three-factor model, excess returns depend on the market, size, and value factors.

Part IV

Applications

Chapter 7

Fixed income basics

This chapter focuses on the basics of fixed income securities, or debt instruments, and interest rates.

7.1 Bond basics

Bonds are the most common type of fixed income securities. They are basically a borrowing arrangement in the form of a security. Firms and governments issue bonds to finance projects and fund operations. The issuer makes two kinds of payments to the bondholders, the investors. When the bond matures, the issuer pays the **par value**, or face value, of the bond. During the life of the bond, the issuer pays the bondholder a periodic **coupon**, typically either annually or semi-annually. The coupon for one bond equals the **coupon rate** times the par value of the bond. Some bonds pay no coupons at all, only the par value at maturity. These are known as **zero-coupon bonds**, or zeros. All the payments, in terms of both size and timing, are fixed when the bond is issued; that is the reason these are called fixed-income securities.

7.2 Bond pricing

Price of any financial asset equals the present value of the future expected cash flows generated by the asset:

$$P = \sum_{t=1}^{\infty} \frac{E\left(CF_t\right)}{(1+r)^t}. \tag{7.1}$$

In the context of bonds, we can simplify this pricing equation. First, bonds have a fixed maturity so we do not need to discount an infinite stream of cash flows. Rather, there will only be T periods of cash flows where T denotes the time to the bond's maturity. Second, since the cash flows are known, we can drop the expectation operator from the numerator. Third, we can separate the stream of coupons and the par value. Doing all these, results in a bond pricing equation

$$P = \sum_{t=1}^{T} \frac{c\,F}{(1+r)^t} + \frac{F}{(1+r)^T}, \tag{7.2}$$

where F is the par value of the bond, c is the coupon rate, and r is the appropriate interest rate to discount the future cash flows. Since c and F are fixed, there is an inverse relation between the bond price P and the interest rate r.

In addition to using Equation 7.2 to find the price of a bond given the interest rate, it can also be used to finding the interest rate given the price. Say, we observe the price of a bond in the market. Now we can use the pricing equation to figure out which interest rate r makes the theoretical price equal to the market price. Such interest rate is known as the **yield to maturity** of the bond.

While the payments of a bond are fixed, there is a risk that the issuer will not have sufficient resources to make the payments when they are due. This is known as **credit risk** or **default risk**. A government issuing bonds in the money it controls has virtually no credit risk. Even if the government does not have enough tax (or other) revenues to pay coupons and par values on its bonds, it can always print more money to solve the problem. Bonds issued by governments in money

they cannot print and, especially, by corporations come with credit risk. A **credit rating** is a measure of the amount of credit risk an issuer, or a bond, carries. Credit risk naturally affects the pricing of the bond. The higher the credit risk, the higher interest rate the investors will apply in discounting the future cash flows from the bond and, hence, the lower the price.

7.3 Term structure of interest rates

So far, we have assumed that the discount rate is the same for all cash flows, no matter when the cash flow occurs. For example, in Equation 7.2 we discount all cash flows from the bond, regardless of their timing, with the same interest rate r.[16] In reality, however, investors may, and typically do, require different interest rates for cash flows happening at different times. Interest rates applied to cash flows in the near future tend to be (although are not always) lower than interest rates applied to cash flow further in the future.

Let us denote by y_t the annualized interest rate, or **yield**, the investors use to discount cash flows occurring t years in the future. A **yield curve** is a graph plotting the yield y_t as a function of time to maturity t. Typically, the yield curve is upward sloping so that the yields at longer maturities are higher than those at shorter maturities. However, this does not have to be the case. The curve can equally well be downward sloping (often called inverted yield curve, long yields are lower than short yields), flat (yields are the same at all maturities), humped (medium-term yields are higher than short and long yields), or any other shape really. The fact that yields vary across maturities is known as the **term structure of interest rates**.

Adjusting the bond pricing equation to account for the term structure is straightforward. Rather than discounting all cash flows with the same interest rate r, we

[16] Cash flows further out in the future do get discounted more as the interest rate is raised to the power of t. However, the annualized interest rate itself is the same for all values of t.

discount cash flows occurring at time t with the matching yield y_t:

$$P = \sum_{t=1}^{T} \frac{c\,F}{(1+y_t)^t} + \frac{F}{(1+y_T)^T}. \tag{7.3}$$

How can we use bond market prices to find the yields for different maturities. Coupon-paying bonds are tricky for this purpose as their prices are made out of cash flows happening at different points in time discounted at different yields. Zero-coupon bonds, however, are perfect for calculating yields. A zero-coupon bond pays a single payment, par value at maturity, making it very easy to back out the yield from the price of the bond. Let us assume that the market price of a zero-coupon bond maturing t years from now is P_t. Now we can use Equation 7.3 to see that the price must satisfy

$$P_t = \frac{F}{(1+y_t)^t}. \tag{7.4}$$

Since we know the par value F, the time to maturity t, and the price P_t, we can rearrange this to solve the unknown yield as

$$y_t = \left(\frac{F}{P_t}\right)^{\frac{1}{t}} - 1. \tag{7.5}$$

We can now use the yields calculated from zero-coupon bonds to price any coupon-paying bond. Let us simplify this calculation by assuming that all bonds, the zero-coupon bonds we use for calculating the yields and the coupon-paying bond we are pricing, have the same par value F. Now we can plug the yields in Equation 7.5 into the bond price in Equation 7.3 and get

$$P = \sum_{t=1}^{T} \frac{c\,F}{\left(1+\left(\frac{F}{P_t}\right)^{\frac{1}{t}}-1\right)^t} + \frac{F}{\left(1+\left(\frac{F}{P_T}\right)^{\frac{1}{T}}-1\right)^T}. \tag{7.6}$$

We can clean this expression significantly. We can rearrange the present value of

the coupon payment at time t as

$$\frac{c\,F}{\left(1+\left(\frac{F}{P_t}\right)^{\frac{1}{t}}-1\right)^t} = \frac{c\,F}{\left(\left(\frac{F}{P_t}\right)^{\frac{1}{t}}\right)^t} = \frac{c\,F}{\frac{F}{P_t}} = c\,P_t, \tag{7.7}$$

and the present value of the par value payment at time T as

$$\frac{F}{\left(1+\left(\frac{F}{P_T}\right)^{\frac{1}{T}}-1\right)^T} = \frac{F}{\left(\left(\frac{F}{P_T}\right)^{\frac{1}{T}}\right)^T} = \frac{F}{\frac{F}{P_T}} = P_T. \tag{7.8}$$

Putting these back into Equation 7.6 gives us the price of the coupon-paying bond as

$$P = \sum_{t=1}^{T} c\,P_t + P_T. \tag{7.9}$$

This means that the price of a coupon-paying bond can be expressed as a weighted sum of prices of zero coupon bonds. For example, a three-year bond paying a 4% annual coupon can be priced as a sum of 0.04 one-year zeros, 0.04 two-year zeros, 0.04 three-year zeros, and one three-year zero. This makes sense, as the combination of the zeros has exactly the same cash flows as the coupon-paying bond. This is an example of pricing financial assets using a **replicating portfolio** approach. We construct a portfolio of zero-coupon bonds whose cash flows exactly match those of a coupon-paying bond, 4% of the par value every year plus the par value at maturity. This is known as a replicating portfolio. Since the two (the replicating portfolio and the bond) have the same cash flows, by **law of one price**, they must have the same price. The replicating portfolio approach is used extensively in pricing derivative securities, which we study in Chapter 10.

7.4 Short rates and forward rates

Let us define some more terminology related to interest rates. As above, y_t is the yield of a t-year zero-coupon bond today. This is also sometimes called the **spot rate**. In finance lingo, spot refers to the price of an asset for immediate delivery. Hence, spot rate refers to the interest rate an investor can earn today on a zero-coupon bond. **Short rate**, r_t, is the one-year interest prevailing at time $t-1$. This is the one-year interest rate an investor will earn buying a one-year bond at time $t-1$ and holding it until time t. For example, r_3 is the one-year interest rate two years from now. Naturally, $r_1 = y_1$.

Let us first start with the unrealistic assumption that we know the future development of short rates. This means that we not only know r_1 for certain but we also know exactly what r_2, r_3, and so on, are going to be. This is not the case in the real world but this simplification helps us learn some important lessons. Later we can relax the assumption and study the more realistic case where future development of short rates is not known.

An investor looking to make an t-year investment has two options. She could invest in a t-year zero-coupon bond. This would earn her a total return of $(1+y_t)^t - 1$. Alternatively, she could invest in a $t-1$-year zero, wait for that to mature in year $t-1$ and then invest in a one-year zero. The total return on this would be $(1+y_{t-1})^{t-1}(1+r_t) - 1$. Since the future short rates known, the total returns on these two alternatives must be identical. Hence, it must hold that

$$(1+y_t)^t - 1 = (1+y_{t-1})^{t-1}(1+r_t) - 1. \tag{7.10}$$

We can rearrange this to solve for the future short rate as

$$1 + r_t = \frac{(1+y_t)^t}{(1+y_{t-1})^{t-1}}. \tag{7.11}$$

It is straightforward to see from Equation 7.11 that an upward (downward) sloping yield curve implies that short rates will be higher (lower) in the future than they are

now. It is important to note that this implication is based on the assumption that future short rates are known with certainty. Since this assumption is unrealistic, we cannot necessarily draw this conclusion from the yield curves in the real world. We discuss this issue further in Section 7.5.

We can now relax the assumption about the future short rates such that we no longer assume that the future short rates are known. This is more realistic and matches the real world experience. Let us also introduce a new financial instrument called a **forward rate agreement**. A forward rate agreement is an agreement about the interest rate to be applied to a loan or a fixed income investment in the future. In a forward rate agreement the two counterparties, the borrower and the lender, agree today what the interest rate will be on a one-year investment to be made on a future date. Let f_t denote the **forward interest rate** from year $t-1$ to t. For example, a borrower can agree with a lender to borrow an amount of money two years from now and pay the money back with interest three years from now. They also agree today that the interest rate the borrower will pay is f_2. Even though the forward rate applies to a transaction to take place in the future, the rate itself is known and agreed upon today.[17]

Now, just like above, an investor is looking to make an t-year investment. She has two options. She could invest in a t-year zero-coupon bond giving her a total return of $(1+y_t)^t - 1$. Alternatively, she could invest in a $t-1$-year zero and agree to a one-year forward rate agreement starting at $t-1$ with a forward rate set to f_t. When her zero-coupon bond matures at time $t-1$ she will invest the money for the final one year at an interest rate she already fixes today using the forward rate agreement. The total return on this would be $(1+y_{t-1})^{t-1}(1+f_t) - 1$. Since the yields and the forward rate are all known today, the total returns on these two alternatives must, again, be identical. Hence, it must hold that

$$(1+y_t)^t - 1 = (1+y_{t-1})^{t-1}(1+f_t) - 1. \tag{7.12}$$

[17] Agreements made today about transactions to take place at a future date are known as derivatives. Chapter 10 introduces the most common types of derivatives and their pricing.

We can rearrange this to solve for the forward rate as

$$1 + f_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}. \tag{7.13}$$

The forward rate must satisfy Equation 7.13 even when we do not know the future short rates. This is because a violation of this equation would result in a simple arbitrage opportunity. An **arbitrage** refers to a transaction that yields an instant profit without any risk of future losses. In a competitive market prices adjust so as to eliminate all arbitrage opportunities. If the forward rate for some reason was greater than implied by Equation 7.13, an arbitrageur would borrow for t years at interest rate y_t, invest the proceeds for $t-1$ years at y_{t-1} and enter a forward rate agreement to invest for one year at time $t-1$ at interest rate f_t. This would give her instant profit as the total return on her borrowing is less than on her investment. If the forward rate for some reason was less than implied by Equation 7.13, the arbitrageur would do the opposite: invest today for t years at interest rate y_t and finance this investment by borrowing for $t-1$ years at y_{t-1} and entering a forward rate agreement to borrow for one more year at time $t-1$ at interest rate f_t. This, again, would result in instant risk-free profit as the interest rate on the investment exceeds that of the borrowing. If either of these arbitrage opportunities arise on the market, traders will exploit them for instant profit, and price will adjust to eliminate the arbitrage opportunity. Hence, Equation 7.13 must hold at all times.

7.5 Theories of term structure

The theories of the term structure of interest rates try to figure out why the yield curve is shaped as it is and why the yield curve is typically upward sloping. We cover two distinct theories here: expectations hypothesis and liquidity preference theory.

The **expectations hypothesis** states quite simply that the forward rates equal the market's expectations of the future short rates, $f_t = E[r_t]$. Plugging this into

Equation 7.13 gives

$$1 + E[r_t] = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}. \tag{7.14}$$

From this we can interpret that, if the expectations hypothesis is correct, an upward (downward) sloping yield curve means that the market expects the short rates to be higher (lower) in the future than they are today. While it would be very tempting to interpret the market interest rates through the lens of the expectations hypothesis, forward rates as expected short rates, the empirical evidence does not support this view. Forward rates are poor predictors of future short rates which is evidence against the expectations hypothesis being the correct framework to interpret the yield curve.

The **liquidity preference theory** takes a completely different view on the yield curve. According to this theory, investors prefer to hold short-term bonds because they are easier and quicker to convert in to cash for full value. An investor is guaranteed to receive the par value of in cash in one year from a one-year bond, but a ten-year bond only guarantees the par value back in ten years. Since investors prefer this liquidity of the short-term bonds over the illiquidity of the long-term bonds, they are willing to pay relatively higher prices for the short-term bonds than for the long-term bonds. This is equivalent to saying that the short-term yields are lower than the long-term yields, or that the yield curve is upward sloping. If the yield curve shape is driven by investors' preference for liquidity, the forward rates are not equal to the expected future short rates as posited by the expectations hypothesis. Indeed, the liquidity preference can generate an upward sloping yield curve even if the short rates are expected to fall in the future.

Summary of Chapter 7

- A bond is a fixed-income security that pays its holders a fixed periodic coupon and, at maturity, its par value. A bond, like any other financial asset, can be priced as the present value of the future payments.
- Yield to maturity is the discount rate that makes the theoretical price of the

bond match the market price of the bond.

- Forward rate is an interest rate applicable to a financial transaction that will take place in the future. The no-arbitrage forward rate can be calculated from two different yields today.

- Term structure of interest rates refers to the fact that investors use different rates to discount cash flows occurring at different points in time. The yield curve is typically, but not always, upward sloping such that the longer-term yields are higher than the shorter-term yields. This could be due to, for example, investors expecting short rates to be higher in the future (expectations hypothesis) or investors preferring the more liquid short-term bonds over the less liquid long-term bonds (liquidity preference theory).

Chapter 8

Fixed income portfolios

This chapter chapter discusses issues central to the management of fixed income portfolios. The focus is on measuring the interest rate risk of bond portfolios and portfolio management approaches.

8.1 Interest rate risk

Since the payments of a bond are fixed, the biggest risk associated with holding a bond is that interest rates rise and the price of the bond falls. Bond investors are very interested in knowing how sensitive the bond prices are to changes in interest rates. Duration and convexity are commonly used measures of the interest rate risk of a bond.

Before looking more closely into the measures of interest rate sensitivity, let us remind ourselves that the price of a bond can be written as the present value of all the payments (coupons and par value) discounted with the yield to maturity of the bond:

$$P = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}, \tag{8.1}$$

where CF_t is the total payment (coupon and par value) in year t and r is the yield

to maturity of the bond. Grouping all the payments together in CF_t makes the formulas below a bit nicer.

Our first, approximate, measure of interest rate sensitivity is the **Macaulay duration**. Mathematically, the Macaulay duration is defined as

$$D = \sum_{t=1}^{T} t\, w_t, \tag{8.2}$$

where

$$w_t = \frac{\frac{CF_t}{(1+r)^t}}{P}. \tag{8.3}$$

While this might look confusing, the Macaulay duration is actually quite intuitive. Let us start from the bottom. The denominator of w_t in Equation 8.3 is simply the price of the bond. The numerator is the present value of of the cash flow occurring at time t. Since the sum of the present values equals the price, w_t can be thought of as a kind of a weight. Every point in time t has a weight w_t which is equal to the relative contribution of the cash flow occurring at that time to the total price of the bond. Over all the values of t, these weights sum up to one. With w_t established as a weight, the Macaulay duration in Equation 8.2 is a weighted average of the times to payments. Each t tells us how long, in years, until that particular payment and w_t weights that time by the relative contribution to the price.

The Macaulay duration D is always less than or equal to the bond's time to maturity T. A zero-coupon bond has only one payment, par value at maturity, whose present value makes up 100% of its price. Hence, its Macaulay duration is simply $D = T \times 1 = T$. A coupon-paying bond makes coupon payments at times $t < T$. These coupons are typically relatively small so that their contribution to the price is also relatively small. However, the weights of these cash flows occurring before maturity is not zero. Hence, the weighted average of times to payment is less than the bond's time to maturity, and $D < T$.

A good question now is why would we care about the Macaulay duration. We care

because it is just one step away from the accurate measure of a bond's interest rate sensitivity. The **modified** duration is defined

$$D^* = \frac{D}{1+r}, \tag{8.4}$$

where D is the Macaulay duration and y is the yield to maturity. Mathematically, the modified duration is negative of the first derivative of the bond price with respect to the yield to maturity divided by the bond price,[18]

$$\frac{1}{P}\frac{\partial P}{\partial r} = -D^*. \tag{8.5}$$

Most importantly, we can use the modified duration to calculate by how much the bond price changes if the yield changes by a small amount. Let us assume that the yield to maturity of a bond rises by Δr. This will result in a percentage drop in the bond price approximately equal to the modified duration times the change in yield. Formally,

$$\frac{\Delta P}{P} \approx -D^* \, \Delta r. \tag{8.6}$$

The price of a bond with a higher modified duration is hence more sensitive to the changes in interest rates. All other things remaining equal a bond with a lower coupon rate will have a higher modified duration. So will a bond with a lower yield to maturity. Also bonds with longer time to maturity tend to have higher modified durations, however, bonds with a low coupon and high yield to maturity are an exception.

Why does the modified duration only provide an approximation of the price change? That is because the bond price is a non-linear function of the yield and the modified duration is a linear approximation. Hence, we should only use it for small values of Δr as the approximation becomes less accurate as the change in yield becomes larger. **Convexity** of a bond measures the second derivative of the bond price with respect to the yield, divided by the bond price. While the exact formula for convexity is beyond the scope of this course, we can use convexity to

[18] Deriving this result is easy but beyond the scope of this course.

provide a more accurate estimate of the percentage change in the bond price as a result of a change in the yield to maturity:

$$\frac{\Delta P}{P} \approx -D^* \, \Delta r + \frac{1}{2} \, \text{Convexity} \, (\Delta r)^2. \tag{8.7}$$

Even with the convexity accounted for, the price change is still an approximation. However, the second-order approximation is good enough such that higher order derivatives are typically ignored even though they are straightforward to calculate.

8.2 Types of bonds

So far we have focused on **straight bonds**, bonds that pay a fixed coupon at regular intervals, and at maturity pay the par value. This, however, is not the only type of bond on the market. Let us look at some of the most common types of more exotic bonds.

A **perpetual bond**, or a perpetuity, is a bond with no maturity date. It simply pays a periodic coupon forever. While nowadays perpetual bonds are a rarity, they are historically somewhat interesting and exhibit some mathematically neat features. The price of a perpetual bond is simply

$$P = \frac{C}{r}, \tag{8.8}$$

where C is the coupon payment, in units of money, and r is the yield. The Modified duration of a perpetual bond is, also, quite simply

$$D^* = \frac{1}{r}. \tag{8.9}$$

A **floating rate bond** is like a straight bond but the coupon rate is not fixed. Rather the coupon rate moves up and down with the market interest rates. Hence, when the interest rates rise (fall) also the coupon rate rises (falls). In the bond pricing Equation 7.2 both the coupon rate c in the numerator and the discount

rate r in the denominator move with market interest rates. This makes the floating rate bond have a very low duration even if it has a long time to maturity.

The payments made by an **indexed bond** are not fixed but tied to a price index (such as consumer price index) or the price of a commodity (such as oil). This is typically done so that the par value of the bond is changed every year by the change in the reference price. Consider, for example, an **inflation-indexed bond**. If inflation in one year is 2%, the par value of the bond is also increased by 2%.

The payments to a **catastrophe bond** depend on whether a pre-determined catastrophe occurs. If no catastrophe occurs, the bond pays a relatively high coupon and par value. If a catastrophe occurs, the coupon and par value payments are either canceled or delayed significantly. This allows the bond issuers to use the catastrophe bonds as an insurance against damages from a catastrophe.

A **callable bond** can be repurchased, or called, at par value by the issuer before its maturity date. The issuer makes the decision to call the bond. The option to call the bond will only be exercised when it benefits the issuer, i.e. when the market value of the bond is higher than its par value. Such call will be costly for the investors and, hence, the coupon rates and yields of callable bonds are typically higher than comparable non-callable straight bonds.

Finally, a **convertible bond** can be converted into a fixed amount of the issuer's common stock. The conversion decision is made by the bondholder who will convert if the stock price is sufficiently high.

8.3 Bond portfolio management

The approaches to managing bond portfolios can be divided in two broad categories: active and passive. Investors with a passive approach to managing their bond portfolio have (at least implicitly) the view that future interest rates cannot be forecasted and that bonds are fairly priced on the market. Hence, passive bond portfolio management focuses on managing risk exposures and investing in bonds

as cost efficiently as possible. Active bond portfolio managers, on the other hand, believe that interest rates can be forecasted or that bonds may be mispriced on the market. Their focus is more on exploiting these market inefficiencies for generating abnormal returns.

A common strategy in passive bond portfolio management is **indexing**. This means investing in a broad bond market index. In effect, the index represents a well-diversified portfolio of bonds. An investor can invest in an index either directly by buying all the bonds in the index, which might be complicated and costly, or indirectly through an index-tracking fund.

Bond prices depend inversely on the market interest rates. When interest rates rise (fall) the value of a bond investors assets, the bonds, decreases (increases). Many bond investors have liabilities that are also sensitive to interest rates. For example, a pension fund's liabilities are the pensions it has promised to pay the policyholders. The pensions can be seen as very long-term fixed income instruments as they are commitments to pay a fixed amount of money very far in the future. A pension fund's problem then is that a drop in interest rates will raise the value of its liabilities by more than it raises the value of its assets. This lowers the net worth of the fund. The solution to such mismatch between the interest rate sensitivity is known as **immunization**. The basic principle of immunization is that the investor tries to match the duration of its assets and liabilities. If the durations of the two sides of the balance sheet match, an small change in the interest rates is going to have the same impact on both sides and the net effect is zero. In principal, a pension fund needs to invest in high-duration, i.e. long-maturity, bonds to match the high duration of its pension liabilities.

Active bond portfolio managers employ a plethora of strategies in pursuit of abnormal returns. Broadly, these can be categorized in four groups, though the classification is by no means exhaustive. A **substitution swap** is a strategy whereby an investor buys one bond that she thinks is underpriced and sells (or shorts) another bond that she thinks is overpriced. The key is that the bonds are as similar as possible (for example similar maturity) but are, for some reason, mispriced relatively to each other. An **intermarket swap** is similar to the

substitution swap but in this case the investor buys and shorts bonds in different sectors of the market, for example in different credit rating categories. The source of abnormal return in both these strategies is the future correction of the initial mispricing.

A **rate anticipation swap** is based on the investor's forecast of future yields. If she expected yields to rise, she buys short-term bonds and shorts long-term bonds. If her yield forecast is correct the long-term bonds will fall in value by more than the short-term bonds, and she earns a positive profit. On the other hand, if she expects the yields to fall, she buys long-term bonds and shorts short term bonds, again earning profit if her forecast turns out to be correct.

Finally, a **pure yield pickup swap** simply takes advantage of the current shape of the yield curve. If the curve is upward sloping the investor buys long-term bonds and shorts short-term bonds. Now she earns a higher yield on the bought bonds than what she has to pay for the shorted bonds. Her hope, of course, is that the yield curve does not flatten which would result in losses. Faced with a downward sloping yield curve, the pure yield pickup swap would prescribe the investors to buy the higher-yielding short-term bonds and short the lower-yielding long-term bonds.

Summary of Chapter 8

- Duration measures the sensitivity of bond price to changes in yields. Convexity measures the non-linearity of the relation between the yield and the bond price.
- In addition to straight bonds, there exist many other, more exotic, types of bonds.
- Bond portfolio management approaches can be broadly defined in passive strategies that attempt to control risk exposures and be cost effective, and active strategies that try to generate abnormal returns.

Chapter 9

Security analysis

This chapter introduces security analysis, especially from the perspective of equity securities. We first quickly review financial statements and then introduce some of the most commonly used methods for valuing a firm's equity.

9.1 Financial statements

There are three main financial statements: income statement, balance sheet, and cash flow statement. An **income statement** is a financial statement that shows the firm's income and expenditures for a given period of time (typically a year or a quarter). It also shows whether the firm is making profit or loss for the given period. The income statement starts with the revenues of the firm and deducts expenses and taxes to arrive at a net income. **Net income** is the profit or loss of the firm. From the net income, the firm pays dividends to its preferred stock and the remained is called its **earnings**.[19] The firm can use the earnings for two purposes. It can either **retain** the earnings in the firm and use them for investing in future growth or it can pay the earnings as **dividends** to common

[19] A preferred stock is a hybrid between equity and debt. It is basically equity but has some priorities over common stock in terms of dividends and treatment in the liquidation of the firm. For our purposes in this course, we can ignore preferred stock.

stock. Typically, firms pay out part of earnings as dividends and retain a part.

A **balance sheet** lists the firm's assets or liabilities at a specific point in time. Assets are what the firm owns and uses to generate income, such as property, plant, equipment, investments, cash, receivables, and so on. Liabilities are what the firm owes, such as debt, payables, taxes, and so on. The difference between the total value of assets and liabilities is known as stockholders' equity.

A **cash flow statement** summarizes the movement of cash and cash equivalents that come in and go out of the firm over a given period of time. A cash flow statement is typically divided into three sources and uses of cash: operating activities, investing activities, and financing activities. It is important to note that the income statement and the cash flow statement measure "flows" over a given period of time, whereas the balance sheet measures "stock" at a given point in time.

From an equity investors perspective, one of the most important accounting-based metrics of a firm's performance is the **return on equity**, ROE. This measure relates the net income of the firm to the stockholders' equity:

$$\text{ROE} = \frac{\text{Net income}}{\text{Stockholders' equity}}. \tag{9.1}$$

Net income is the amount of money the firm generates that the stockholders can freely decide what to do with. Stockholders' equity is, in effect, an accounting measure of the amount of money the stockholders have invested in the firm. Hence, ROE is a measure of how much profit the investment in the firm is generating for the stockholders relative to their investment.[20]

The ROE can be broken down in three components using a so-called **DuPont decomposition**:

$$\text{ROE} = \frac{\text{Net income}}{\text{Equity}} = \frac{\text{Net income}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}. \tag{9.2}$$

The first term, net profit margin, measures the profitability and operating effi-

[20] This is different from the stock return that uses market prices, rather than accounting values.

ciency of the firm. The second, asset turnover ratio, measures the efficiency of the firm's use of its asset. The third term, equity multiplier, measures the firm's financial leverage.

9.2 Equity valuation

Equity valuation is the process of finding a subjective fair value of a firm's equity or stock. This value may well differ from the market price. We first pick a valuation model, then make some assumptions, plug those assumptions into the model, and get an equity value as the output. The output can at best be as good as the inputs. Hence, if we make garbage assumptions we are going to get a garbage valuation.

Equity valuation can be used as the basis of investment decisions. If the market price of a stock is lower (higher) than an investor's subjective valuation, she should buy (sell) the stock. Firms can use equity valuation for their own investment, capital budgeting, and merger and acquisition decisions. Working through equity valuation models is also quite interesting on its own right. Below we introduce three different valuation models: comparables, dividend discount models, and free cash flow models.

9.3 Method of comparables

The point of the method of comparables (or comps) approach to valuation is to calculate **valuation multiples** for a group of comparable firms and base the valuation on the average of this group.

Valuation multiples are scaled measures of the firm value. They take a measure of the firm's market value and divide that by some accounting-based measure of the firm's size, such as earnings, cash flow, or book equity. Probably the most commonly used valuation multiple is the price-to-earnings ratio which is the firm's share price divided by the expected earnings per share next year. Another common

price ratios include the price-to-dividend ratio, which divides the share price by the expected dividend, and price-to-book, which is the market value of equity divided by the book value of equity.

The process of valuing firm X using the comparables approach is the following:

- Define the comparable firms. These should be firms that are sufficiently similar to X in terms of their business.
- Choose the appropriate valuation multiple. The valuation multiple is defined as a market value divided by a fundamental (accounting) value.
- Calculate valuation multiples for the comparable firms and firm X.
- Calculate the average of the comparable firms' valuation multiples.
- Multiply the average valuation multiple by firm X's fundamental value. This result in the equity price firm X would have if it had the same valuation multiple as the comparable firms on average.

It is worth noting that the method of comparables is not, strictly speaking, a valuation model. Rather it is a relative pricing model.

9.4 Dividend discount model

To motivate the dividend discount model, assume an investor who buys a stock today with the intention to sell after one year. One year from today, she will receive a dividend and then sell the stock. Today, she does not know what the dividend will be, but her expected value of the dividend is $E[D_1]$. Also, she does not know what price she will get when she sells the stock, but the expected price is $E[P_1]$. The equity cost of capital, and hence the expected return of the stock, is equal to k.[21] The value of the stock for our investor today, V_0, is present value

[21] Assume the investor has estimated k, for example, using the CAPM.

of her expected cash flow next year:

$$V_0 = \frac{E[D_1] + E[P_1]}{1+k}. \tag{9.3}$$

The stock value today is the sum of next year's expected dividend and expected stock price discounted. But what about $E[P_1]$, the expected price next year? We can use the same logic and write the next year's price as the discounted sum of the expected dividend and expected price two years from now:

$$E[P_1] = V_1 = \frac{E[D_2] + E[P_2]}{1+k} \tag{9.4}$$

We can plug this into Equation 9.3 and get

$$V_0 = \frac{E[D_1] + \frac{E[D_2]+E[P_2]}{1+k}}{1+k}, \tag{9.5}$$

which we can rearrange as

$$V_0 = \frac{E[D_1]}{1+k} + \frac{E[D_2]}{(1+k)^2} + \frac{E[P_2]}{(1+k)^2}. \tag{9.6}$$

We can continue the process of replacing the expected price at time t with the discounted expected dividend and price at time $t+1$. Continuing this infinitely yields the price of the stock today as the present value of an infinite stream of dividends:

$$V_0 = \sum_{t=1}^{\infty} \frac{E[D_t]}{(1+k)^t}. \tag{9.7}$$

This is the **dividend discount model**, or DDM. In order to value equity using the DDM, we need to forecast the firm's future dividends and discount them using the firm's equity cost of capital.

An interesting observation arises from the logic above. Even when the investor only plans to hold the stock for one year, she cares about all the dividends the firm will ever generate. This is because the price at which she will sell depends on

all the future dividends. Hence, a rational stock investor should never be myopic, no matter how short her intended holding period is.

Obviously we cannot forecast expected dividends all the way to infinity. In order to operationalize Equation 9.7, we need to make an assumption regarding the dividends. The easiest application of the DDM is to assume that dividends grow at a constant rate g. This means that the expectation of next year's dividend is $E[D_1]$, the expectation of the year 2 dividend is $E[D_2] = (1+g)E[D_1]$, the expectation of the year 3 dividend is $E[D_3] = (1+g)^2 E[D_1]$, and so on. Using Equation A.10 from Appendix A yields

$$V_0 = \frac{E[D_1]}{k-g}. \tag{9.8}$$

This is known as the **constant dividend growth model** or the **Gordon model**. There are a few things worth noting in Equation 9.8. First, the numerator is the expected dividend next year. It is a common mistake to put this year's realized dividend, D_0, in the numerator. Note that under the assumption of constant dividend growth, $E[D_1] = (1+g)D_0$.

Second, the expected growth rate of the dividends needs to be strictly less than the discount rate, $g < k$. Otherwise the price coming out of the formula makes no sense.

Third, we can rearrange Equation 9.8 as

$$k = \frac{E[D_1]}{P_0} + g, \tag{9.9}$$

which gives us an estimate of the expected stock return in the case where the market expects a constant dividend growth rate. The expected return equals the sum of the dividend yield (dividend divided by price) and the dividend growth rate.

What is the rate of dividend growth? To answer this question, let us follow a simple logic. First, the firm generates earnings that it can either pay out as dividends or

invest in new projects. The **payout rate** is the percentage of the earnings paid out as dividends while the **plowback rate** is the percentage of earnings invested. Naturally, plowback rate = 1 − payout rate. Hence, we define new investment as

$$\text{New investment} = \text{Earnings} \times \text{Plowback rate}. \tag{9.10}$$

The new investment will result in increased future earnings. More specifically, the increase in future earnings is equal to the amount invested times the firm's return on equity:

$$\text{Change in earnings} = \text{New investment} \times \text{Return on equity} \tag{9.11}$$

The earnings growth rate is simply the change in earnings divided by the level of earnings. Combining this with Equations 9.10 and 9.11 gives the earnings growth rate as

$$\text{Earnings growth rate} = \frac{\text{Change in earnings}}{\text{Earnings}} \tag{9.12}$$

$$= \text{Plowback rate} \times \text{Return on equity}. \tag{9.13}$$

If we assume that the firm will maintain a constant dividend payout rate, the growth rate of dividends matches the growth rate of earnings. Hence, the growth rate of dividends is given by

$$g = \text{Plowback rate} \times \text{Return on equity}. \tag{9.14}$$

Equation 9.14 shows that a higher return on the new investment results in a higher dividend growth rate. Also, a higher plowback rate results in higher dividend growth, as long as the return on the new investment is positive. Of course, increasing the plowback rate lowers the next year's expected dividend in Equation 9.8.

9.5 Free cash flow model

The discounted free cash flow model starts from the notion that the two sides of the balance sheet must match. On the left-hand side of the balance sheet we have the value of the firm's business, called enterprise value in this context. On the right-hand side we have the value of debt and the value of equity. The two sides must match:

$$\text{Enterprise value} = \text{Debt value} + \text{Equity value}. \tag{9.15}$$

From this we can solve for the equity value as

$$\text{Equity value} = \text{Enterprise value} - \text{Debt value}. \tag{9.16}$$

This is the basic idea of the free cash flow model valuation of equity. Rather than valuing equity as the present value of dividends, it values equity as the enterprise value minus debt. Finding the enterprise value is less complicated than forecasting dividends, as we do not need to make assumptions regrading the firm's payout decisions. The enterprise value is simply the present value of free cash flows to the firm, appropriately discounted. Now we just need answers to two questions. First, how to calculate free cash flow to the firm? And second, what is the appropriate discount rate?

The **free cash flow to firm**, FCFF, is defined as the firm's total cash flow that can be used as payments to debt and investors and as investments to generate future growth. Formally, FCFF, is calculated as

$$\text{FCFF} = \text{EBIT}(1-\tau) + \text{Depreciation} - \text{CapEx} - \text{Change in NWC}, \tag{9.17}$$

where EBIT is the earnings before interest and taxes, τ is the corporate tax rate, depreciation is the reduction in value of assets from wear and tear, CapEx is the capital expenditure, and NWC stands for net working capital, the difference between short-term assets and short-term liabilities. That answers our first question.

Since we are discounting cash flows to the whole firm, we should discount them

by a discount rate that applies to the whole firm, not just its equity. The firm is financed by equity and debt. A fraction w_E of the financing is in equity with a discount rate k_E. The remainder, fraction $w_D = 1 - w_E$, is financed with debt with a discount rate k_D. Since the payments to debt are tax-deductible, the cost of the debt for the firm is only $(1-\tau)k_D$ rather than the investors' return requirement k_D. The firm's weighted average cost of capital is given by

$$WACC = w_E\, k_E + w_D(1-\tau)k_D. \tag{9.18}$$

Now we can just put these two together to solve the enterprise value as

$$\text{Enterprise value} = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+WACC)^t} \tag{9.19}$$

and the equity value as

$$\text{Equity value} = \text{Enterprise value} - \text{Debt value}. \tag{9.20}$$

The most difficult part of using the free cash flow model is forecasting the expected free cash flow to firm. This step requires forecasting future sales, operating expenses, taxes, capital requirements, and all other relevant factors of the whole firm. A thorough understanding of the firm and its operating environment is required. On the other hand this step is very interesting as it necessarily leads to an in-depth understanding of the firms operations.

Stock investors are sometimes accused of being short-sighted and only narrowly interested in profit maximization. However, equity valuation is anything but short-sighted and narrow. First, we already saw above how an investor who only plans to hold a stock for a short period of time cares about the full future of the firm's free cash flows. Second, equity is a residual claim on the firm's cash flow; stockholders only receive cash from the firm after all other stakeholders (suppliers, employees, tax authorities, debtors, and so on) have received their payments. Hence, to understand the value of a firm's equity, we need to understand everything about

the firm. Equity valuation, contrary to some accusations, is extremely broad and has a very long time perspective.

Summary of Chapter 9

- There are three main financial statements. An income statement tracks a firm's income and expenditures over a period of time, a balance sheet lists a firm's assets and liabilities at a point in time, and a cash flow statement tracks the flow of cash in and out of the firm over a period of time.
- The method of comparables finds the market value of equity the firm would have if it had a valuation multiple equal to the average of the comparable peer firms.
- The dividend discount model finds the value of equity as the present value of future dividends, discounted using the firm's equity cost of capital.
- The free cash flow model finds the value of equity as enterprise value minus debt. The enterprise value is the present value of the firm's free cash flows, discounted using the firm's weighted average cost of capital, WACC.

Chapter 10

Derivative securities

This chapter introduces derivative securities, or derivatives, financial instruments whose value depends upon the value of other assets. We first look at some of the most common types of derivatives: forwards, futures, and options. Then we discuss the pricing of derivatives as well as different option trading strategies.

10.1 Forwards

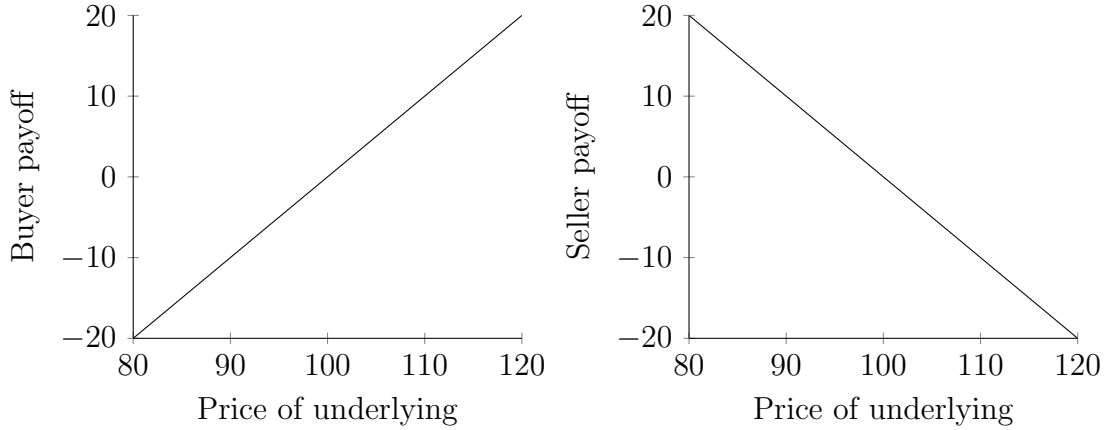
Fundamentally, a forward is simply an agreement of a transaction to take place at a future date. The forward agreement specifies at least the following:

- The asset to be transacted, known as the **underlying asset**. Theoretically, this could be anything. In practice, derivatives are written on stocks, bonds, indexes, currencies, interest rates, commodities, other derivatives, and many others.
- **Expiration date**. This is the future date when the transaction in the underlying asset takes place.
- **Forward price**, F. This is the price at which the underlying asset is transacted on the expiration date.

There are two counterparties in a forward agreement: buyer and seller. The buyer is said to have a **long position**. She buys the underlying asset for the forward price on the expiration date. The seller is said to have a **short position**. She sells the underlying asset for the forward price on the expiration date.

Figure 10.1 presents the **payoff diagrams** of a forward contract with forward price equal to 100. A payoff diagram plots the profit or loss of a forward counterparty as a function of the price of the underlying asset at the time when the contract expires. If the price of the underlying ends up being equal to the forward price, 100, both counterparties have a zero payoff. If the price of the underlying is 110, the forward buyer makes a profit of 10. This is because she can buy the underlying asset for 100 from the forward seller and sell the asset on the spot market for 110. This results in a profit of 10. In this situation the forward seller will lose 10 as she has to sell the asset that is worth 110 for the forward price of 100. In contrast, if the price of the underlying ends up being 90, the forward buyer will lose 10 and forward seller will gain 10.

Figure 10.1: Payoffs of forward buyer and forward seller.



It is obvious from Figure 10.1 that the forward payoff diagrams are **linear**. This is why forwards are sometimes called linear derivatives. Mathematically, the payoff of the forward buyer is equal to $P_T - F$, where P_T is the price of the underlying asset at the time when the forward contract matures. The forward seller's payoff is given by $F - P_T$. It is useful to note that the sum of the two counterparties' payoffs is zero.

10.2 Futures

Fundamentally, a futures contract is identical to a forward contract: an agreement to trade a certain quantity of an underlying asset at a future date for a pre-specified price. There are, however, some key differences between forwards and futures. The important differences are listed below.

- Futures are traded on **exchanges** whereas forwards are privately negotiated between the two counterparties, typically a broker-dealer and its customer. Hence, forwards are said to trade **over-the-counter** (or OTC).
- Futures are highly **standardized** whereas forwards are not. Because futures are traded on exchanges, they need to be standardized in terms of underlying assets (type, quality, quantity) and delivery dates. Forwards, as private contracts between the two counterparties, do not need to be standardized and can have very specific features.
- Profits and losses on forwards occur at maturity, as discussed above. Futures, on the other hand, are **marked-to-market**. Marking-to-market means that a futures trader's **margin account** at the futures exchange is credited (debited) every day for the gains (losses) of her position in the futures contract. The eventual profit or loss for similar forward and futures will be the same. The difference is that for the forward, the profit or loss occurs at maturity wheres for the futures it accrues over the life of the contract.

10.3 Options

Options differ crucially from linear derivatives. Forwards and futures are agreements of future transactions that are binding for both counterparties. An option gives its **holder** the right, but not the obligation, to buy or sell the underlying asset in the future for a pre-determined price. The option holder's counterparty is known as an option **writer**. The option writer has the obligation to sell (buy)

the underlying asset if the holder wishes to exercise her option to buy (sell). The fact that the holder has a right but no obligation makes options **non-linear** and a bit more complicated than the linear derivatives.

There are two kinds of options. A **call option** gives its holder the right to buy the underlying asset in the future for a pre-determined price, known as **strike price**. If the holder wishes to buy, the option writer must sell the underlying asset for the strike price. A **put option** gives its holder the right to sell the underlying asset for the strike price. Again, if the holder wishes to sell, the writer is obliged to buy.

Quite obviously, the holder is never going to exercise her option if that would result in losses for her. For example, a holder should never exercise a call option to buy a stock for 100 if the market price is less than 100. She should only exercise the option if the stock price is over 100, when she gets a positive payoff. This means that the option holder's payoff at the option expiration will always be non-negative. This further implies that the writer's payoff will always be non-positive. Hence, in order to acquire the option in the first place from the writer, the holder must pay the writer a price for the option. This price is known as the option **premium**.

Figure 10.2 presents the payoff diagrams of a call option holder and writer. The option has strike price equal to 100 and premium equal to 4.[22] The solid line plots the payoffs at option maturity ignoring the premium paid. The dashed line plots the profit and loss taking the premium into account.

If the price of the underlying is 110, the option holder can buy the underlying asset for 100 from the option writer and sell the asset on the spot market for 110. This results in a payoff of 10. However, she paid a premium of 4 to get the option in the first place and her total profit is 6. If the price of the underlying ends up being 90, the call holder does not exercise her option and her payoff is 0. However, she paid 4 for the option, resulting in a total loss of 4. The option writer's payoffs, profits, and losses are naturally a mirror image of those of the option holder. It is easy to see from the payoff diagrams why options are sometimes called **non-linear**

[22]Here we take the premium as given. Below we study how to calculate the appropriate premium.

Figure 10.2: Payoffs of call option holder and writer.

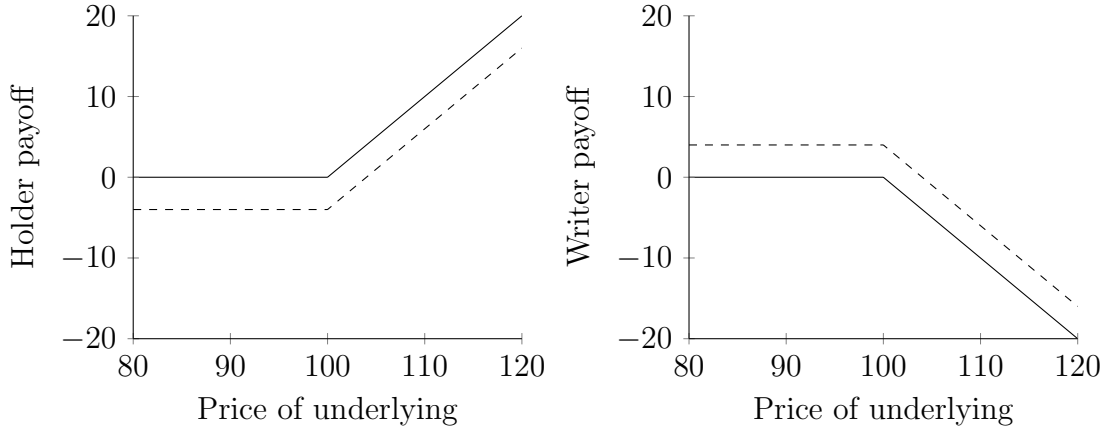
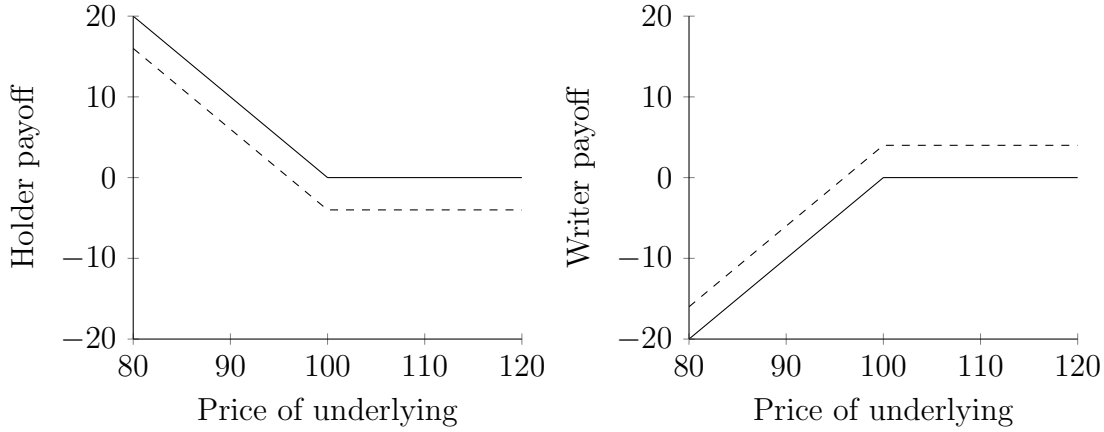


Figure 10.3: Payoffs of put option holder and writer.



derivatives.

Figure 10.3 plots the payoff diagrams for put options. A put option holder benefits from a decrease in the price of the underlying asset. If the price is lower than the strike price, she can buy the underlying asset on the market for the low price and sell it to the option writer for a higher price, resulting in a profit. If the price is higher than the strike, she will not exercise her option. The option writer's payoffs are, again, a mirror image of the holder's payoffs.

It is important to note that there are four distinct payoff diagrams: a holder and a writer of a put and of a call. It is a common mistake to think that a put option

holder has the same payoff as a call option writer.

Table 10.1 summarizes the payoffs of different option counterparties mathematically.

Table 10.1: Option payoffs.

Counterparty	Call	Put
Holder	$\max(P_T - X, 0)$	$\max(X - P_T, 0)$
Writer	$-\max(P_T - X, 0)$	$-\max(X - P_T, 0)$

10.4 Pricing forwards

Let us now turn our attention to the pricing of derivative securities. Our focus here is on finding the fair price for the derivative. In the case of forwards, this boils down to finding the appropriate forward price. In the case of options, we want to find the correct premium.

Let us start with forwards. In a forward contract, neither of the counterparties pays the other when the contract is agreed. Rather, the forward price F is set so that the contract is satisfactory to both counterparties. Pricing a forward contract is then about finding such F. Let us denote the time to expiration by T, the current price of the underlying asset by P_0, and the risk-free rate of interest by r_f.

The forward buyer agrees to buy the underlying asset at the expiration date for price F. Alternatively, she could borrow P_0 today, use the borrowed money to buy the underlying asset, and eventually repay the loan when time T has passed. By then, the borrowed amount has grown to $P_0(1 + r_f)^T$. Hence, the buyer would agree to the forward contract only if $F \leq P_0(1 + r_f)^T$.

In contrast, the forward seller agrees to sell the underlying asset for price F at the expiration date. Alternatively, she could sell the asset today for P_0, deposit the proceeds, and withdraw the deposit after time T has passed. By then the deposit is worth $P_0(1 + r_f)^T$. Hence, the seller would agree to the forward contract only if

$F \geq P_0 (1 + r_f)^T$.

Combining the two conditions for the forward price, we can see that there is only one price to which both counterparties can agree. Hence, in an equilibrium, the forward price must be

$$F = P_0 (1 + r_f)^T. \tag{10.1}$$

There is an even stronger argument for why this is the only possible forward price. Let us assume that for some reason, someone is offering to enter a forward contract for which $F > P_0 (1 + r_F)^T$. Now, an **arbitrageur** can sell the forward and make instantaneous risk-free profit. This is known as arbitrage, and should not happen in a competitive equilibrium. As said, the arbitrageur sells the forward because it is offered at too high a price. At the same time, she borrows the present value of the forward price, $F/(1 + r_f)^T$ and uses P_0 to buy the underlying asset. When the forward contract expires, she sells the underlying asset to the forward buyer for F. By that time, her debt will have grown to F, so she uses the proceeds from selling the underlying asset to repay the debt. Hence, her net cash flow at the contract expiry is zero. Today, however, she is left with with a net cash flow of $F/(1 + r_f)^T - P_0$. Because $F > P_0 (1 + r_f)^T$ the arbitrageur's net cash flow today is positive. In total, her portfolio yields a sure instant profit without risk of future losses.

If someone is offering the forward contract with $F < P_0 (1 + r_f)^T$, the arbitrageur would buy the forward, sell the underlying asset, and deposit the present value of F. At contract expiration, she would withdraw the deposit and use the proceeds to buy the underlying asset from the forward seller. Again, she would make instant profits with no risk of future losses.

A portfolio which yields instant profit with no risk of future losses is know as an **arbitrage**. Quite obviously, arbitrage should not happen in a competitive equilibrium. As arbitrage is possible both when $F < P_0 (1 + r_f)^T$ and when $F > P_0 (1 + r_f)^T$, the only price that prevents arbitrage is $F = P_0 (1 + r_f)^T$. Hence, this is the only forward price that can prevail in an equilibrium.

This is an example of **no-arbitrage pricing** of derivatives. The point of this approach is simply to find which price for the derivative contract prevents arbitrage.

Interestingly, and importantly, the no-arbitrage forward price does not depend on the counterparties' expectations about the price of the underlying asset at the forward contract's expiration.

10.5 Pricing options

Option pricing is the process of finding the appropriate premium at which it makes sense for the writer to write the option and for the buyer to buy the option. Like above with forwards, we use the no-arbitrage principle to find the option premium. The basic idea is that we can construct today a portfolio of the option's underlying asset and the risk-free asset that has the same payoffs as the option at maturity. Since our **replicating portfolio** has the same payoff as the option, its price today must match the option premium today. Otherwise arbitrage would be possible.

Unlike forwards, however, options cannot be replicated **statically**. On the contrary, a **dynamic** replication strategy is required to match the non-linear payoff function of an option. This means that the replicating portfolio needs to be adjusted continuously by buying or selling the underlying asset and selling or buying the risk-free asset. Moreover, the appropriate amounts of the assets to be held by the replicating portfolio depend on the assumptions regarding the development of the price of the underlying asset.

The **Black-Scholes model**[23] assumes that the price of the underlying behaves like a geometric Brownian motion. With this assumption, the Black-Scholes premium of a call option is given by

$$V_C = P_0 \mathcal{N}(d_1) - X\, e^{-r_f T} \mathcal{N}(d_2), \tag{10.2}$$

[23] Fischer Black and Myron Scholes, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy*, 1973.

where

$$d_1 = \frac{\ln\left(\frac{P_0}{X}\right) + \left(r_f + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \tag{10.3}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}. \tag{10.4}$$

In these formulas $\mathcal{N}$ is the cumulative standard normal distribution and σ is the standard deviation of the returns of the underlying asset. While the formulas look complicated, using them is straightforward. We just plug in the known values for the current price of the underlying (P_0), option strike price (X), option time to maturity (T), standard deviation of the return of the underlying (σ), and risk-free rate of return (r_f).

The Black-Scholes formula for the put option premium is

$$V_P = X\, e^{-r_f T}\mathcal{N}(-d_2) - P_0\,\mathcal{N}(-d_1), \tag{10.5}$$

where d_1 and d_2 are as above.

An important observation arises from the Black-Scholes option pricing formulas: expected return does not enter the formulas. The riskiness (σ) of the underlying asset matters for the pricing of options, but not the expected return of the underlying asset. Conversely, this means that option prices are not informative about the expected returns of the underlying asset. They are, however, informative about the volatility.

10.6 Option strategies

Options can be used in portfolio management either together with the underlying asset or without the underlying asset. Here, we introduce some of the most common option strategies.

A **protective put** is an options trading strategy used by investors to protect an existing investment position against downside risk. It involves the purchase

of put options on an asset the investor already owns. The primary purpose of a protective put is to provide an insurance against a significant drop in the value of the underlying asset. If the price of the underlying asset drops below the strike price, the put option provides positive payoffs limiting the downside risk of the combined portfolio of underlying and put. This insurance, however, is not without a cost as the investor has to pay the premium on the put option. A **covered call** is a strategy employed by investors who own an underlying asset, typically stocks, and want to generate additional income while potentially limiting their upside profit potential. It involves selling call options on an asset the investor already holds. The investor earns the premium on the call option but faces a limited upside potential.

Options are particularly well-suited for investing in volatility. An investor may have the view that volatility will increase in the future, i.e. that changes in asset prices, up or down, will become larger. To benefit from this view she can buy a call option and a put option. The combined portfolio of options will result in a positive payoff, net of premiums, if the price of the underlying asset ends up moving a lot up or down from its current value before the options expire. If the strike prices of the call and the put are the same, this position is know as a **straddle**. If the strike prices differ, then the position is called a **strangle**. If the investor expects volatility to be low, and asset price movements to be very small, she can write a call and a put. Such a strategy is known as a **short straddle** or **short strangle**.

Options can also be used to speculate on the direction of the moves of the underlying asset. For example, an investor expecting the underlying asset to appreciate, can buy a call option. If she finds the premium to be too high, she can finance the purchase of the call by writing another call option with a higher strike price. This would be known as a **bull spread**. Conversely, an investor expecting the underlying asset to depreciate, can enter a **bear spread** by buying a put option and writing another put option with a lower strike price.

Chapter 11

Active portfolio management

This chapter discusses active portfolio management, the process of deviating from benchmark portfolio weights in order to obtain higher risk-adjusted returns.

11.1 Active and passive portfolio management

Investors' approaches to portfolio management can be divided in two broad categories, active and passive. In the **passive** approach, the investor's goal is to replicate the returns of a market portfolio. She is not trying to a earn higher returns by picking assets that are likely to perform best. Rather, she focuses on replicating the market portfolio, the value-weighted portfolio of all risky assets, as cost efficiently as possible. The passive approach is optimal if markets are efficient or if the investor knows that she has no informational advantages over the other market participants. The investor can either replicate the market portfolio directly by holding individual assets in the market portfolio weights or she can invest in an index tracking fund that does the allocation on her behalf.

An investor following the **active** approach to portfolio management deviates from the market portfolio weights in order to outperform the market on a risk-adjusted basis. The focus of an active portfolio manager is to find underpriced assets to

overweight and overpriced assets to underweight. The success of active management requires that markets are inefficient to certain extent and that the investor has some informational advantage over the market as a whole.

Note that the dichotomy between active and passive management is not related to the frequency of trades. A passive investor trades very little because she does not need to trade to maintain the passive portfolio. An active investor may trade a lot or not. An investor that only holds shares of Apple, and never trades, is an active investor as her portfolio deviates massively from a market portfolio.

11.2 Active management on aggregate

The market portfolio is the aggregate portfolio of all investors combined. Let us assume that we start from an idealistic situation where all investors are passive and hold the market portfolio. That is, all investors' risky asset portfolio is exactly the same. Now, one investor, let us call her investor A, decides to become active and deviate from the market portfolio weights. To do this, she sells some of her holdings in asset 1 and buys more of asset 2. She does this because she believes asset 1 is overvalued compared to asset 2. But who does she sell asset 1 to? And who does she buy asset 2 from? In order for investor A to complete these trades, there must be another investor, call her investor B, who buys asset 1 and sells asset 2. Now, compared to the market portfolio, investor A is underweighting asset 1 and overweighting asset 2. She is clearly active. But so is investor B; she is overweighting asset 1 and underweighting asset 2. Investor B's active deviations from the market portfolio are exactly the opposite to investor A's active deviations. This holds true more broadly. Whenever we aggregate over all the active investors, their active deviations from the market portfolio cancel out. As an aggregate, the active investors hold the market portfolio.

As time goes by, either asset 1 or asset 2 will end up having a higher return. Depending on which has a higher return, either investor A or B will end up beating the market, i.e. having a higher return than the market portfolio. However, the

other active investor will lose to the market portfolio by exactly the same amount as the other one beats the market by. Because the active investors, on aggregate, hold the market portfolio, their aggregate return must also be the same as the market portfolio return. One will have higher, one will have lower. This also holds more broadly, any higher-than-market returns for some active investors must be offset by lower-than-market returns for some other active investors. On aggregate, active investment does not generate positive, or negative, abnormal returns. If there are costs associated with active management, as there usually are, the active investors must on aggregate lose to the market portfolio by the amount of their costs.

11.3 Theory of active management

Say an investor wants to be active and has views about the expected returns of some assets. How should she allocate her portfolio? To answer this question, we follow the model of Treynor and Black.[24] Let us assume that asset excess returns are represented by a single-index model:

$$R_{i,t} = \alpha_i + \beta_i \, R_{M,t} + e_{i,t}. \tag{11.1}$$

If an investor believes that all alphas are zero, she should opt for a passive strategy of investing her wealth in the value-weighted market portfolio. If, however, she believes that some alphas are different from zero, she should actively deviate from the market portfolio weights by overweighting the positive-alpha assets and underweighting the negative-alpha ones. But by how much should she over- and underweight different assets? The Treynor-Black model provides a simple framework for working out the optimal weights.

The active investor should invest in two distinct portfolios. First, some fraction of her wealth should be invested in the market portfolio M, a value-weighted portfolio

[24] Jack L. Treynor and Fischer Black, "How to Use Security Analysis to Improve Portfolio Selection," *Journal of Business*, 1973.

of all risky assets. Second, the remainder of her wealth should be invested in the active portfolio A that contains the over- and underweights in the non-zero-alpha assets. Hence, the investor's total portfolio P is a combination of portfolios M and A. Being rational, the investor wishes to maximize the Sharpe ratio of the total portfolio. To find the investor's optimal allocation, we need to solve two problems. First, we find the optimal allocation between market portfolio and the active portfolio. Then, we find the optimal composition of the active portfolio.

The investor should invest fraction w_A^* in the active portfolio and fraction $1 - w_A^*$ in the market portfolio. The optimal active portfolio weight is given by

$$w_A^* = \frac{w_A^0}{1 + (1 - \beta_A)\, w_A^0}, \tag{11.2}$$

where

$$w_A^0 = \frac{\frac{\alpha_A}{\sigma_{e_A}^2}}{\frac{\mu_M}{\sigma_M^2}}, \tag{11.3}$$

β_A is the beta of the active portfolio, α_A is the alpha of the active portfolio, $\sigma_{e_A}^2$ is the idiosyncratic variance of the active portfolio, μ_M is the expected return of the market portfolio, and σ_M^2 is the variance of the market portfolio. What do these formulas do? Equation 11.3 says that if the beta of the active portfolio is one, then the investor should invest in the active portfolio an amount that is proportional to the active portfolio's reward-to-variablility ratio relative to the market portfolio's reward-to-variability ratio. Equation 11.2 makes a small adjustment to this weight to account for the fact that the active portfolio beta may differ from one.

With the optimal allocation between the M and A portfolios solved, we can turn our attention to the optimal composition of the active portfolio A. To solve this, it is useful to calculate the Sharpe ratio of the investor's total portfolio P. The squared Sharpe ratio of the P portfolio is given by

$$S_P^2 = S_M^2 + \left(\frac{\alpha_A}{\sigma_{e_A}}\right)^2, \tag{11.4}$$

where S_M^2 is the squared Sharpe ratio of the market portfolio. Since the Sharpe

ratio is positive, and the investor takes the market portfolio and its Sharpe ratio as given, maximizing the Sharpe ratio of P boils down to maximizing the the ratio $\frac{\alpha_A}{\sigma_{e_A}}$. This ratio of alpha to idiosyncratic volatility is known as **information ratio**. We could intuitively think that in order to maximize the Sharpe ratio of the total portfolio, the investor should maximize the Sharpe ratio of the active portfolio. This, however, is not the case, as we can see in Equation 11.4. In order to maximize the Sharpe ratio of P, the investor has to maximize the information ratio of A.

How should the investor construct portfolio A so as to maximize its information ratio? The optimal allocation, within portfolio A, to asset i is given by

$$w_i^* = \frac{w_i^0}{\sum_{j=1}^N w_j^0}, \tag{11.5}$$

where

$$w_i^0 = \frac{\alpha_i}{\sigma_{e_i}^2}. \tag{11.6}$$

The optimal active portfolio invests in the non-zero-alpha assets in weights that are proportional to their ratio of alpha to idiosyncratic variance. An investor applying the Treynor-Black model should proceed as follows:

1. Determine the alphas, betas, and idiosyncratic volatilities of the non-zero-alpha assets.
2. Solve for the optimal composition of portfolio A.
3. Calculate the alpha, beta, and idiosyncratic volatility of portfolio A (using the formulas in Section 5.4) and the expected return and volatility of portfolio M.
4. Solve for the optimal allocation between the M and A portfolios.

11.4 Performance evaluation

The goal of active portfolio management is to generate risk-adjusted returns that are higher than those of the market portfolio. Hence, it makes sense to evaluate the realized performance of and active investor to see if she has been successful or not. This section introduces some of the metrics most commonly used in investment performance measurement and evaluation.

Say we wish to evaluate the past performance of portfolio P. We have data on the returns of the portfolio, as well as the market portfolio and a risk-free asset. The easiest measure of performance is the simple average return,

$$\bar{r}_P = \frac{1}{T}\sum_{t=1}^{T} r_{P,t}. \tag{11.7}$$

Evaluating performance based on the simple average is, however, far from anything feasible as the measure ignores the risks taken to achieve the returns.

One easy way to adjust the portfolio performance for risk is to use the empirical **Sharpe ratio**,

$$S_P = \frac{\bar{r}_P - \bar{r}_f}{\sigma_P}, \tag{11.8}$$

where $\bar{r}_f$ is the average risk-free rate of return and σ_P is the return standard deviation of portfolio P. Generally speaking, an optimal portfolio is the one that maximizes the Sharpe ratio. Hence, the empirical Sharpe ratio is the theoretically correct measure of risk-adjusted performance for a total portfolio of risky assets. A potential downside is that the Sharpe ratio is heavily affected by market returns. When market returns are high (low), the portfolio returns and Sharpe ratios tend to also be high (low).

Jensen's alpha is a way to adjust for the market return. Jensen's alpha is simply the intercept term of single-index model and can be defined as

$$\alpha_P = \bar{r}_P - \left[\bar{r}_f + \beta_P\left(\bar{r}_M - \bar{r}_f\right)\right], \tag{11.9}$$

where β_P is beta of portfolio P. The term in the square brackets is the expected return of portfolio P under the single-index model assuming that the portfolio has a zero alpha. The difference between the average return and the zero-alpha-expectation is then the portfolio's estimated alpha. Jensen's alpha can be easily extended to a multifactor framework by adding other relevant systematic risk factors such as size and value in the Fama-French three-factor model. The single-index model alpha is widely used in practice, but as a shortcoming does not adjust for the idiosyncratic risk.

Adjusting for the idiosyncratic risk of an actively managed portfolio may be important. While the market portfolio only contains systematic risk, an actively-managed portfolio that deviates from the market portfolio may contain significant idiosyncratic risk. Also, the Treynor-Black model shows that investors should maximize the ratio of alpha to idisyncratic volatility when optimizing the actively-managed part of their portfolio. Hence, our final performance metric is the empirical **information ratio**,

$$IR_P = \frac{\alpha_P}{\sigma_{e_P}}, \tag{11.10}$$

where σ_{e_P} is the idiosyncratic volatility of portfolio P. This would be the theoretically correct performance metric when assessing the performance of the active part of an investor's portfolio (in the context where the investor invests in the market portfolio and an active tilt portfolio like in the Treynor-Black model).

Interpretation of empirically estimated alphas, and information ratios, is plagued by the difficulty of separating **luck and skill**. A positive alpha is commonly taken as evidence that the portfolio manager has some skill in generating high returns. However, in the estimation of the alpha, it is assumed that the average idiosyncratic return during the sample period is zero. This however, might not be the case. The portfolio manager might just have been lucky and experienced, on average, positive idiosyncratic returns. But since in the estimation the average idiosyncratic return is assumed to be zero, all these lucky returns contribute to the alpha. In an infinitely long sample period the average idiosyncratic return will converge to zero. But we do not have infinitely long histories of returns. Typically we work with just a few years of data, much less than infinity. In a finite sample,

separating luck and skill is very difficult.

11.5 Summary of Chapter 11

- Active portfolio management refers to deviations from the market portfolio, not the frequency of trading.
- On aggregate, active management cannot produce returns higher than the market portfolio.
- The Treynor-Black model provides the optimal portfolio allocation for an active investor. Most interestingly, to maximize the Sharpe ratio of her total portfolio, the investor should maximize the information ratio of her active deviations from the market portfolio.
- Theoretically, the correct performance metric for a total portfolio is the Sharpe ratio, whereas the performance of an active portfolio should be assessed using the information ratio. Jensen's alpha is used commonly in performance evaluation as well.
- Disentangling luck and skill in performance evaluation is very difficult.

Chapter 12

Sustainable investing

In recent years, sustainable investing has received a lot of attention. This chapter provides a brief overview of sustainable investment practices, approaches, and their desired effects.

12.1 Between traditional investing and philanthropy

In the traditional approach to investments, investors choose portfolios to maximize the expected returns while not taking excessive risks. They view assets as something that generate expected returns and are exposed to systematic and idiosyncratic risks. The focus is fully in the financial aspects. At the other extreme, a philanthropist spends her money on donations with the hope of having a maximal positive societal impact in the chosen area without expecting to receive anything in return. Sustainable investing is somewhere between traditional investing and philanthropy.

Sustainable investing refers to a range of practices in which investors aim to achieve financial returns while promoting long-term environmental or social value. An

investor following a sustainable approach, combines traditional investment approaches with environmental, social, and corporate governance (ESG) insights. A key goal of sustainable investing is to ensure that firms are not judged solely on financial gains but on a broader picture of what and how they contribute to the society. Sustainable investment philosophy makes investors think critically about their investments' potential impacts as they relate to environmental, political, and societal landscapes.

An investment's sustainability is often evaluated based on the ESG factors. These factors are:

- **E**nvironmental: A firm's impact on the environment, such as its carbon footprint, waste, water use and conservation, and the clean technology it uses and creates in its supply chain.
- **S**ocial: A firm's impact on society and how it advocates for social good and change in key social issues such as human rights, racial diversity within hiring and inclusion programs, employees' health and safety, and community engagement.
- **G**overnance: How a firm is managed, or governed, for driving positive change, including quality of its management and board of directors, executive compensation and diversity, shareholder rights, overall transparency and disclosure, anti-corruption, and corporate political contributions.

12.2 Approaches

There are different approaches to incorporating sustainable aspects to portfolio management. On the level of a diversified portfolio, the most common ones are negative and positive screenings. A **negative screening** refers to the practice of excluding specific assets (for example stocks of firms) because of their particularly poor ESG performance. A **positive screening** refers to only investing in assets with particularly good ESG performance.

On the level of a more concentrated portfolio, sustainable investment strategies also include activist investing and impact investing. **Activist investing** is the process of buying a significant equity stake in a firm for the purpose of changing how it operates. **Impact investing** refers to a targeted investment aimed at solving a specific social or environmental problem. It includes community investing, where capital is directed to traditionally underserved individuals or communities and financing is provided to businesses with clear social or environmental purposes.

12.3 Effects

How does sustainable investing actually promote sustainability? It is quite clear that targeted impact investment can have a direct impact by allocating more resources to solving social and environmental problems. Also activist investors can directly change how their target firms operate. But what about the screening approaches, especially the negative screening most likely to be applied by most investors?

Negative and positive screening increase the demand for securities issued by sustainable firms and decrease the demand for securities issued by unsustainable firms. If such practices become common enough, they are going to affect asset prices. Stocks of sustainable firms will become more expensive and stocks of unsustainable firms will become cheaper. This has an impact on the firms' cost of capital. The cost of capital will become lower (higher) for the sustainable (unsustainable) firms. This should make investing more in sustainability easier for the sustainable firms and provide incentives for the unsustainable firms to improve their sustainability. Since firms cater to investors' preferences, more focus on ESG issues by investors should drive firms to focus on these issues as well.

Summary of Chapter 12

- Sustainable investing broadens investors' focus from purely financial objectives to environmental, social, and governance impact of their investments.
- Negative and positive screening limit the universe of investable assets. Activist investing attempts to directly affect the way a firm operates. Impact investing allocates resources directly to solving a particular social or environmental problem.
- Sustainable investing can contribute positively either directly by allocating resources to solving problems or changing the way firms operate, or indirectly through asset prices.

Part V

Appendix

Appendix A

Time value of money

This chapter provides a brief overview of time value of money.

A.1 Money at different times

Almost any financial asset, or investment, generates cash flows that happen at different points in time. Typically, an investor pays for the asset when acquiring it and later receives cash flows from the asset in the form of interest, dividend, or proceeds from selling the asset.

The crucial issue here is that the cash flows happening at different points in time cannot be directly compared. The investor cannot simply calculate the total net cash flow generated by an asset and invest if this total is greater than zero. This is because an amount of money today is more valuable than the same amount in the future. In order to compare cash flows happening at different times, we must **discount** them to present day values. In essence, discounting finds the **present value** of a future cash flow, i.e. how much the cash flow is worth in present day money.

A.2 Interest rates

Why is money today worth more than money in the future? This is the same as asking why are interest rates positive. The answer lies in **impatience**, **risk**, and **inflation**. First of all, people are impatient with their consumption. Given the opportunity to consume the same basket of goods either today or next year, most would choose to consume today. To move consumption to the future, they would require more consumption goods in the future than they would receive today. If consumers are indifferent between consuming one basket today and consuming $1+r^*$ baskets next year, this implies that the **real interest rate** is r^*. If consumers are impatient with their consumption, the real interest rate is positive.

Second, there may be risks associated with moving consumption to the future. Let us ignore impatience for a while and assume that a consumer is willing to change one basket of consumption today for one basket of consumption next year. If there is no risk associated with receiving the consumption basket next year, the real interest rate would be zero. However, if there is a risk that she will not receive the consumption basket (or that she will receive a less-than-full basket) next year, the consumer will require a positive real interest rate as compensation for bearing this risk. This part of the interest rate is called a **risk premium**.

Third, price levels tend to rise over time, i.e. inflation tends to be positive. So far we have discussed real interest rates, interest rates in terms of consumption baskets. However, in the real world we typically operate with **nominal interest rates**, interest rates in terms of units of money. If a consumer is willing to exchange one basket of consumption today for one basket of consumption next year (implying that the real interest rate is zero) she would require a positive nominal interest rate if she expects the price of the consumption basket to increase. To move her consumption to the next year, the consumer can sell a consumption basket today, deposit the proceeds in a savings account for a year, and next year withdraw the deposit, with interest, and use this money to purchase a consumption basket. If she expects the price of the consumption basket to increase over the next year by i, she would require a nominal interest rate of i on the savings account to be able

to purchase the consumption basket next year.

Together these three things, impatience, risk, and inflation, make interest rates positive and a unit of money today more valuable than the same unit of money in the future.

A.3 Discounting

How can we compare cash flows happening at different points in time? The simple answer is, by calculating the values of the cash flows in present day money and comparing these present values. This is what **discounting** is all about.

To understand how discounting works, assume that an investor invests amount PV in a bank account that pays interest rate r. For example, if the interest rate is 5%, $r = 0.05$. In one year's time the account balance will include the original deposit of PV and the accrued interest of $PV \times r$. Hence, in one year, the deposit of PV has grown to $PV(1 + r)$. In two years' time the balance will include PV, two years' interest on PV equal to $2 \times PV \times r$, and the interest on the first year's interest of $PV \times r \times r$. Hence, the total balance after two years is equal to $PV(1 + 2r + r^2) = PV(1 + r)^2$. Every year the account balance grows by a factor $(1+r)$, and after t years, the balance will be $PV(1+r)^t$. Hence, the **future value** in year t of PV can be expressed as

$$FV_t = PV(1 + r)^t. \tag{A.1}$$

Discounting finds the **present value**, the value today, of a future amount of money. As the future value in one year of depositing PV at interest rate r equals $FV_1 = PV(1+r)$, dividing both sides of the equation by $(1+r)$ yields the present value of receiving FV_1 next year as:

$$PV = \frac{FV_1}{1 + r}. \tag{A.2}$$

Similarly, as the future value of PV in two years' time is $FV_2 = PV(1+r)^2$, the present value of receiving FV_2 in two years must be

$$PV = \frac{FV_2}{(1+r)^2}. \tag{A.3}$$

Using the same logic, dividing both sides of equation A.1 by $(1+r)^t$, the present value of receiving FV_t in t years is

$$PV = \frac{FV_t}{(1+r)^t}. \tag{A.4}$$

Equation A.4 can be used to calculate the present value of any future amount of money.

Financial assets typically generate **cash flows over multiple years**. The big question is, of course, what is the total present value of such asset. In other words, what is the total present value of all the cash flows the asset generates. The answer is surprisingly simple. The total present value of a stream of cash flows is just the sum of the present values of all the future cash flows.[25] The present values can be summed (or subtracted) because they are all expressed in the same unit, today's money. The present value of receiving a cash flow CF_1 in year 1, CF_2 in year 2, CF_3 in year 3, and so on until a final cash flow CF_T in year T is simply the sum of the present values:

$$PV = \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots + \frac{CF_T}{(1+r)^T}. \tag{A.5}$$

Identically, this can be written as a sum operation:

$$PV = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}. \tag{A.6}$$

[25] We use the term "future value" above to motivate the present value formula. Hereafter, we refer to the future amounts of money as "cash flows". This is more consistent with the goal of valuing assets whose purpose is to generate cash flows in the future.

The present value formulas for annuities and perpetuities are useful in valuing assets that generate cash flows far into the future. An **annuity** is a series of fixed cash flows happening at a specified frequency over the course of a fixed time period. The present value of receiving cash flow CF every year for T years is, naturally, equal to

$$PV = \sum_{t=1}^{T} \frac{CF}{(1+r)^t} = CF \sum_{t=1}^{T} \frac{1}{(1+r)^t}. \tag{A.7}$$

The sum term has a simpler expression and the whole present value can be written as

$$PV = CF \left[\frac{1-(1+r)^{-T}}{r} \right]. \tag{A.8}$$

While this formula might not seem too intuitive, it has the benefit of not containing the sum operator. This formula is very useful when valuing an asset that generates a constant cash flow annually for a very long, but finite, time period. Note that there is no need for a time subscript in CF as the cash flow is the same every year.

A **perpetuity** is an infinite series of periodic cash flows. In essence, a perpetuity is an annuity with $T = \infty$. Increasing T to infinity in Equation A.8 yields the present value of a perpetuity as

$$PV = \frac{CF}{r}. \tag{A.9}$$

This formula is, of course, useful in valuing assets that generate a constant perpetual cash flow.

A **growing perpetuity** is an infinite series of cash flows that grows at a constant rate. In the first year, the cash flow is CF_1. In the second year, the cash flow is $CF_1(1+g)$, where g is the annual growth rate of the cash flow. The cash flow is $CF_1(1+g)^2$ in the third year, $CF_1(1+g)^3$ in the fourth year, and so on. A little bit of mathematical magic yields the present value of such a growing perpetuity as

$$PV = \frac{CF_1}{r-g}. \tag{A.10}$$

This formula is useful in valuing assets that generate a perpetual cash flow that grows at a constant rate. As an example, the dividends paid by a firm to its shareholders can be modeled as a growing perpetuity. There are two important things to note about this formula. First, the discount rate needs to be greater than the growth rate of cash flows, $r > g$. If this is not the case, the formula produces a nonsensical present value. Second, the numerator is the cash flow in the first year, i.e. one year from now. Also note that assuming a zero growth rate ($g = 0$) naturally makes Equation A.10 identical to Equation A.9.

A.4 Net present value

Equation A.6 presents the formula for calculating the present value of receiving a stream of cash flows in the future. Such a stream of cash flows can be considered a financial asset. But what about the cost of obtaining the asset in the first place? **Net present value**, or **NPV** is the answer to the question "What is the net value of acquiring an asset, that generates a stream of stream of cash flows, for a given price?" NPV is simply the present value of all cash flows, positive and negative, associated with the acquisition of the asset. Formally the NPV is equal to

$$NPV = CF_0 + \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots + \frac{CF_T}{(1+r)^T}, \tag{A.11}$$

where CF_0 is simply the cash flow today. This can also be written as a sum operation

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}. \tag{A.12}$$

Note that since $(1+r)^0 = 1$,

$$\frac{CF_0}{(1+r)^0} = CF_0. \tag{A.13}$$

In the simplest case CF_0 represents the negative cash flow that the investor must pay today to acquire the asset that has positive future cash flows. But this does

not have to be the case. The cash flows can be negative for multiple years in the beginning of an asset's life. This is the case, for example, if an investor in a startup company is expected to make multiple investments in the early stages of the company's life.

A.5 Infinitely lived stream of cash flows

A financial asset may be expected to have an infinite life, i.e. it is expected to generate cash flows forever. In such a case, finding the NPV of the asset involves discounting an infinite stream of cash flows. This, of course, can only be done with the help of a simplifying assumption.

The most common assumption is that an asset generates varying cash flows between today and year T, and after year T the cash flows grow at a constant rate g. In this case, the cash flow in year $T+1$ is CF_{T+1}, the cash flow in year $T+2$ is $CF_{T+1}(1+g)$, cash flow in year $T+3$ is $CF_{T+1}(1+g)^2$ and so on. We can use Equation A.10 to find the value of the constantly growing cash flows, happening after year T, in year T as

$$V_T = \frac{CF_{T+1}}{r-g}. \tag{A.14}$$

Note that this value is not measured in today's money but in year T's money. It still needs to be discounted, divided by $(1+r)^T$, in order to calculate the present value in today's money.

Combining the present values of all cash flows between years 0 and T and the present value of the constantly growing cash flows happening after year T yields the NPV of the infinitely lived asset:

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t} + \frac{1}{(1+r)^T} \frac{CF_{T+1}}{r-g}. \tag{A.15}$$

A.6 Risky cash flows

Above, we assumed that the future cash flows are known. This, however, is typically not the case, but there is some uncertainty, or risk, associated with the cash flows. Such risk is easily incorporated into the present value calculation. First, the numerator of the formula should contain the **expected** cash flow. This is the most likely value of the future cash flow.[26] Second, the discount rate r should take the **riskiness** of the cash flow into account. In short, the riskier the cash flow, the higher the discount rate. This makes sense intuitively, as greater risk then results in lower present value of the cash flow.

Formally, the NPV of an asset with risky cash flows is

$$NPV = \sum_{t=0}^{T} \frac{E\left[CF_t\right]}{(1+r)^t}. \tag{A.16}$$

Summary of Chapter A

- Money today is worth more than the same amount of money in the future. Hence, money at different times cannot be directly compared.
- Comparison of money at different times is facilitated by calculating the present value of future money. This is also known as discounting.
- NPV is the present value of all cash flows associated with an asset.
- NPV can be used to calculate the financial value of any stream of cash flows: a stock, a bond, a loan, a derivative, a firm as a whole, an individual asset of a firm, or any other contract or asset that results in cash flows.

[26] See Appendix B for a primer on random variables and a more detailed explanation of expected values of random variables.

Appendix B

Random variables

This chapter provides a short primer on random variables. Students with less familiarity with random variables may find this a useful background for some of the Chapters above.

B.1 Random variables

A random variable is a variable whose eventual value is subject to uncertainty. A random variable has multiple (at least two) potential values, only one of which will be realized. Once the value is realized, no uncertainty remains.[27]

In the context of this booklet, the two most important random variables are cash flows and returns. Future cash flows and returns have not yet been realized, so there is uncertainty associated with them. However, historical cash flows and returns are already realized, and hence there is no uncertainty associated with them anymore.

There are two main types of random variables:

[27] A coin toss provides a useful example. As long as the coin is not tossed the outcome can either be heads or tails. Once the coin has landed, however, there is no uncertainty as the outcome of the toss has been realized.

- A **discrete** random variable has a finite number of potential values. Examples include a coin toss (2 possible values: heads and tails) and dice roll (typically six possible values). In both of these cases, it possible to list all the potential outcomes because there is a finite number of them.

- A **continuous** random variable has an infinite number of potential values. An example is temperature tomorrow at noon. If the temperature could be measured infinitely accurately, there would be an infinite amount of potential outcomes of the random temperature.[28] The most important random variables in this booklet, cash flows and returns, can be thought of as continuous random variables.

It is important to note that randomness does not necessarily imply that all potential outcomes of a variable would be equally likely. In a fair coin toss heads and tails are equally likely, but tomorrow's temperature is very likely closer to today's temperature than 10 degrees warmer than today. A **probability distribution** is a function that describes the probability of each potential value of a random variable. In the case of a discrete variable, the probability distribution is either a list of the potential values and their probabilities, or a mathematical function mapping potential values of the random variable with probabilities. In the case of continuous variables, the probability distribution is always a mathematical function.

To establish notation, denote the random variable by X. The potential values of a discrete random variable are listed by x_i with associated probabilities given by p_i. Note that the sum of the probabilities must equal 1. For example, in a roll of a dice there are 6 potential values: $x_1 = 1$, $x_2 = 2$, $x_3 = 3$, $x_4 = 4$, $x = 5$, and $x_6 = 6$ and the probability of each value is the same $p_1 = p_2 = \ldots p_6 = \frac{1}{6}$.

[28] Note that the set of potential values for a continuous random variable does not necessarily extend from negative infinity to positive infinity. For example, it is physically impossible for the temperature tomorrow at noon to be less than $-273.15°$C.

B.2 Expected value, variance, and covariance

Expected value measures the "most likely" outcome of a random variable.[29] A bit more formally, expected value is the average value of the random variable in a very large sample of repetitions. For example, a single roll of a dice can yield any value between 1 and 6. When very many dice are rolled, the average of the dice will be 3.5. This is the expected value of rolling a dice.

The expected value of a random variable X is denoted by $E[X]$. For a discrete random variable, the expected value can be calculated as the probability-weighted sum of the potential values:[30]

$$E[X] = \sum_i p_i \, x_i. \tag{B.1}$$

The following two properties of expected values are useful in understanding the other chapters of this booklet.[31] First, multiplying X by a constant a also multiplies the expected value by a:

$$E[a\,X] = a\,E[X]. \tag{B.2}$$

Second, the expected value of the sum of two random variables (X and Y) is equal to the sum of the expected values of the two random variables:

$$E[X+Y] = E[X] + E[Y]. \tag{B.3}$$

[29] Strictly speaking, expected value is exactly equal to the most likely value only if the distribution function is symmetric.

[30] For a continuous random variable the expected value is given by the definite integral

$$E[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx,$$

where $f(x)$ is the probability distribution function of X. For the purpose of understanding the content of this booklet, it is sufficient to understand expected value of discrete random variables as given in Equation B.1.

[31] These properties hold for discrete and continuous random variables.

Putting these two properties together yields the expected value of a weighted sum of two random variables as

$$E\left[a\,X + b\,Y\right] = a\,E\left[X\right] + b\,E\left[Y\right], \tag{B.4}$$

where a and b are constants. How does this relate to finance? If X and Y are the random future returns of two assets and a and b their respective weights in a portfolio, this formula gives the expected return of the portfolio.

Variance measures how far the potential values of the random variable are spread out from the expected value. For discrete random variables, the variance can be calculated as the probability-weighted sum of the potential values' squared deviations from the expected value:[32]

$$Var[X] = \sum_i p_i\,\left[x_i - E(X)\right]^2. \tag{B.5}$$

Oftentimes, especially in finance, the variation of a random variable is measured by **standard deviation** rather than variance. Standard deviation is simply the square root of variance:

$$SD[X] = \sqrt{Var[X]}. \tag{B.6}$$

Like expected return, variance also has some simple properties useful for finance applications.[33] First, multiplying X by a constant a multiplies the variance by a^2:

$$Var[a\,X] = a^2\,Var[X], \tag{B.7}$$

which implies that

$$SD[a\,X] = a\,SD[X]. \tag{B.8}$$

[32]For a continuous random variable the expected value is given by the definite integral

$$Var[X] = \int_{-\infty}^{\infty} \left[x - E(X)\right]^2 f(x)\,dx.$$

For the purpose of understanding the content of this booklet, it is sufficient to understand variance of discrete random variables as given in Equation B.6.

[33]Again, these properties hold for discrete and continuous random variables.

Second, the variance of the sum of two random variables (X and Y) is equal to the sum of the variances plus two times their covariance:

$$Var[X+Y] = Var[X] + Var[Y] + 2\,Cov[X,Y]. \tag{B.9}$$

Covariance measures how strongly two random variables are related. A large positive covarinace implies that the two variables are likely to move in the same direction, i.e. when one has a high realization, the other is likely to have a high realization as well. Conversely, a large negative covarinace implies that the two variables are likely to move in the opposite directions, i.e. when one has a high realization, the other is likely to have a low realization.

Covariance can be calculated as the **correlation** between the two random variables times the product of the variables' standard deviations:

$$Cov[X,Y] = Cor[X,Y] \times SD[X] \times SD[Y]. \tag{B.10}$$

Correlation is a standardized measure of the dependence between two random variables. It can take values between -1 and 1. A correlation of 0 means that the two random variables are independent of each other. A correlation of 1 implies a perfect positive dependence and that the variables always move in the same direction. A correlation of -1 implies a perfect negative dependence and that the variables always move in opposite directions. A correlation between 0 and 1 (-1 and 0) implies that the two variables tend to move in the same (opposite) direction. Correlation between asset returns is fundamental to portfolio diversification.

Covariance also has its own scaling rule. Multiplying X by a and Y by b multiplies the covariance by the product of a and b:

$$Cov[a\,X, b\,Y] = a\,b\,Cov[X,Y]. \tag{B.11}$$

Putting B.7, B.9, and B.11 together yields the variance of a weighted sum of two

random variables as

$$Var[a\,X + b\,Y] = a^2\,Var[X] + b^2\,Var[Y] + 2\,a\,b\,Cov[X, Y]. \tag{B.12}$$

The covariance in this equation can of course be written out as in B.10. How does this relate to finance? If X and Y are the random future returns of two assets and a and b their respective weights in a portfolio, then this formula gives the variance of the portfolio return.

Sometimes Greek letters are used to denote the statistics related to random variables. In the most common notation, μ denotes expected value, σ denotes standard deviation, and ρ denotes correlation.

B.3 Linear regression

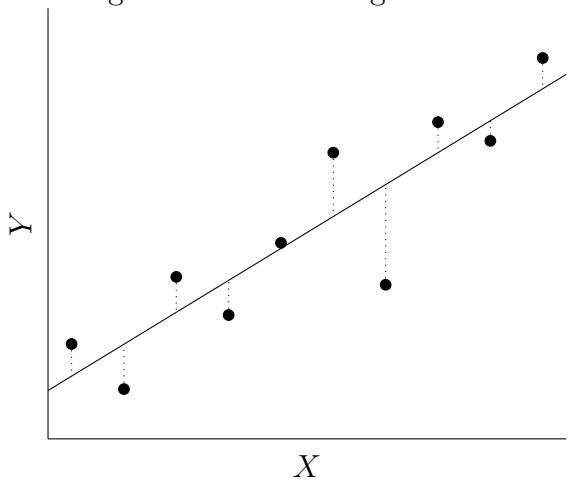
Linear regression is useful for describing the relation between two or more variables. Figure B.1 provides an illustration of a **simple linear regression**. There are two variables, X and Y. There are 10 observations for which the value of both variables are recorded. Each observation is represented by a dot in figure. It is clear from just looking at the figure that there is a positive relation between X and Y: when X is larger, also Y tends to be larger. However, the relation does not seem to be perfect and the observations do not all fall on a straight line or any other simple function.

In regression terminology, Y is often called the dependent variable while X is called the independent or explanatory variable. A linear regression simply fits a straight line through the observations so that the line does the best possible job in describing how Y depends on X. This is represented by the solid line in the figure. The regression model takes the form

$$Y_i = \alpha + \beta\,X_i + \varepsilon_i, \tag{B.13}$$

where i indexes the observations and α (Greek letter alpha) and β (Greek letter

Figure B.1: Linear regression.



beta) are the coefficients to be estimated. The final term in the regression model, ε_i, is called an error term, or residual, and represents vertical distance of observation i from the linear line. The dotted lines in Figure B.1 plot the error terms related to each observation.

Estimating the regression boils down to finding coefficients α and β that best describe the relation between X and Y. Coefficient α is the intercept and tells at which point the regression line crosses the Y axis. Coefficient β tells the slope of the regression line. A common method for estimating the coefficients is the ordinary least squares or OLS. This method prescribes to find such α and β that the sum of squared error terms is minimized. The OLS estimates for the coefficients are given by

$$\beta = \frac{Cov[Y, X]}{Var[X]} \tag{B.14}$$

$$\alpha = \bar{Y} - \beta\bar{X}, \tag{B.15}$$

where $Cov[X, Y]$ covariance between Y and X, $Var[X]$ is the variance of X, and $\bar{Y}$ and $\bar{X}$ are the averages of Y and X, respectively. Typically, we would use a statistical software to estimate the regression coefficients.

In a **multiple linear regression** there are multiple explanatory variables. An

example of a multiple linear regression with two explanatory variables would have the form

$$Y_i = \alpha + \beta X_i + \gamma Z_i + \varepsilon_i. \tag{B.16}$$

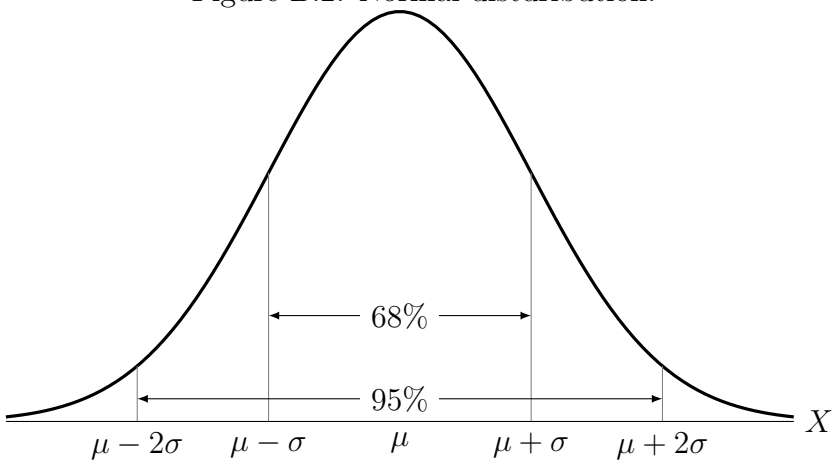
Compared to the simple linear regression in Equation B.13, here Y also depends on variable Z with a slope coefficient γ. The OLS estimates for the coefficients of the multiple linear regression are quite messy and have to be estimated using a statistical software.

Linear regressions have many useful applications in finance. Most importantly in the context of this course, the simple linear regression model can be used to estimate beta, the measure of systematic risk under the capital asset pricing model, CAPM. The CAPM beta is equal to the slope coefficient in a simple linear regression where the dependent variable is the excess return of an asset and the explanatory variable is the excess return of the market portfolio. Another application of linear regressions in the context of this course is the estimation of various factor models. For example, we can use factor models to assess the performance of an active investment fund. We can regress the excess returns of the fund on systematic risk factors and hence decompose the fund's returns into components due to systematic risk exposures (represented by the slope coefficients), idiosyncratic risk exposures (represented by the error terms), and the fund manager's skill (represented by the intercept term). Due to the α often being used to denote the regression intercept term, the return of a fund or an investment strategy not explained by risk exposures is often called alpha.

B.4 Normal distribution

A very commonly used continuous probability distribution is the **normal** (or Gaussian) distribution. The normal distribution has many positive attributes. First, it is fully described with just two parameters: expected value and variance. Second, it is relatively easy to work with mathematically. Last, but not least, many real-world random variables, such as financial returns, seem to empirically have

Figure B.2: Normal disturibution.



probability distributions that closely resemble the normal distribution. Figure B.2 plots the probability density function of a normally distributed random variable.

The probability function of the normal distribution peaks at the expected value, denoted by μ in the figure. One of the nice features of the normal distribution is that 68% of realizations lie within plus-minus one standard deviation from the expected value and 95% within plus-minus two standard deviations. For example, assume that annual returns of a stock are normally distributed with expected value 10% and standard deviation 20%. Then there is a 68% probability that next year's return well be between -10% and 30%, and a 95% probability that it will be between -30% and 50%.

Summary of Appendix B

- Random variable is a variable whose eventual value is currently subject to uncertainty.
- Expected value measures the most likely outcome of a random variable.
- Variance and standard deviation measure the dispersion of the possible outcomes around the expected value.

- Correlation and covariance measure the degree of dependence between two random variables.
- The normal distribution is a useful model for the distribution of a random variable.
- In the context of this course, future asset returns are a random variable, expected return is the expected value of the future return, variance and standard deviation are used as measures of risk, correlation and covariance play a pivotal role in diversification, and normal distribution is a useful model for future returns.

THE END